Supplement to NeST-Com: Fixed-Budget Next-Scale Token Communication

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Implementation recipe, complete results, paired statistics and source examples. Numerical tables use the same frozen CSV sources as the main manuscript.

S1 Reproduction configuration

NeST-Com transmits source-token prefixes from a shared, frozen next-scale representation. The complete machine-readable configuration, candidate pools, PHY layouts, model manifest and reference vectors are supplied in `inputs/reproduction/`. The repository entry `experiments/nestcom-v9-completion-20261101/verify_cdf_ac.py` replays the integer coding checks on CPU; `verify_assets.py` resolves an external asset directory and checks file identities. No private server path is required for those verification commands.

S1.1 Representation, models and numerical route

The tokenizer has a 4096-entry codebook and ten maps of side lengths $(1, 2, 3, 4, 5, 6, 8, 10, 13, 16)$. The VAR-d16 prior uses null class 1000 and greedy prediction, without CFG. Complete coarse maps determine every position's probability in the next map. Both terminals use those known maps; no unreceived same-scale token enters the receiver's entropy context.

The adapted decoder checkpoint is update0038000. It minimizes MSE+0.1 LPIPS on 20 000 ImageNet training images with a 50/25/25 mixture of continuous latents, fully quantized latents, and interpolated prefix-completed latents. The last input is $F_b + \alpha(F - F_b)$, $\alpha \sim U[0, 1]$, with F_b completed from eight source scales. Selection minimizes full-calibration mixture image loss. Tokenizer, prior and codebook remain frozen. The recipe is `experiments/var-latent-enhancement-20260917/configs/experiment.json`.

The NeST reference route is Python 3.10.12, NumPy 2.2.6, Torch 2.11.0+cu128, CUDA 12.8 and cuDNN 91900. Inference is FP32, batch 1, eval/no-grad, TF32 off, matmul precision highest, deterministic algorithms and cuDNN deterministic on, benchmark off. The revised timing uses two intra-op and two inter-op threads on two distinct physical CPU cores; S7 separates it from the historical SMT-pair run. Device PCI/UUID, tensor-state hashes, checkpoint-file hashes and source-file hashes are separate manifest fields. Reproducing a floating-point probability stream requires this numerical route as well as the weights.

Ada retains LlamaGen-L/VQ16 (upstream commit `ce98ec41803a74a90ce68c40ababa9eaeffeb4ec`), predicted ConvNeXt-tiny classes, conditional-branch entropy CDFs and CFG 4/T1/top-k2000/top-p1 sampled completion. Its 10-bit class is paid. The public LlamaGen weight URLs and revision are included in the manifest. ARPC uses its public 0.06M preset, CFG 3, temperature 0.5 and seed 0; its paid caption and source container are preserved. Latent JSCC uses MSE+0.1 LPIPS+0.01 latent consistency and shared D_c, with 100k updates; SwinJSCC uses RGB MSE, 200k updates and minimum calibration-MSE selection. Both use the fixed 20k training subset. External models retain their own numerical routes rather than inheriting NeST's environment.

Public VAR/LlamaGen/ConvNeXt/ARPC sources, exact file pins, available tensor-state pins and architecture configurations are listed in `model_manifest.json`. Project-trained D_c and JSCC checkpoints are retained in the project storage; no public download URL is asserted for them. Their recipes and exact checkpoint identities are supplied. Set `NESTCOM_ASSET_ROOT`, or provide `NESTCOM_ASSETS_JSON` mapping checkpoint names to local paths. A file SHA is never presented as a tensor-state SHA.

S1.2 Integer probabilities and independently terminated streams

FP32 logits pass through `torch.log_softmax`; NumPy float64 then performs stable normalization. For vocabulary $V = 4096$ and total frequency $M = 2^{24}$,

$$p_j = \frac{\exp(\ell_j - \max_k \ell_k)}{\sum_k \exp(\ell_k - \max_k \ell_k)}, \qquad f_j = \lfloor p_j(M - V) \rfloor + 1,$$

$$d = M - \sum_j f_j, \qquad f_{j^*} \leftarrow f_{j^*} + d, \quad j^* = \arg\max_j p_j.$$

Exact ties use the smallest codebook index. The int64 CDF is $[0, \mathrm{cumsum}(f)]$, with strictly positive frequencies. No CDF table is transmitted. The implementation is `src/var_comm/entropy.py`.

The arithmetic interval starts at $[0, 2^{32} - 1]$. An incremental encoder evaluates successive prefixes; each is finalized independently, without changing the incremental state. Finalization emits $b = 0$ if low$< 2^{30}$, otherwise $b = 1$, followed by pending+1 inverse bits. RX initializes a 32-bit state, uses bounded zero extension, and requires exactly 30 terminal zero reads. It re-encodes recovered tokens and requires exact equality to the received payload. The paid (m, K) specifies the token count. Prefix streams are not slices of a longer codeword. Missing files and invalid model/CDF inputs are execution errors, not receiver successes or NULLs.

S1.3 Digital frame and rejection rules

The 68-symbol header carries a 12-bit profile, CRC16 and six zero termination bits. Its convolutional encoder has constraint length 7 and octal generators $(171, 133)$, rate $1/2$; its 68 coded bits are repeated to 136 bits and mapped to 68 rotated-BPSK symbols. Terminated Viterbi decoding requires final state zero. CRC uses polynomial 0x1021, initial 0xFFFF, MSB-first integer fields. The body CRC covers the 13-bit length, source bits and known zero padding.

For $q = \log_2 M_{\mathrm{QAM}}$, $n = (N - 68)q$, $k = \lfloor rn \rfloor$ and source capacity $k - 29$. All 12 declared MCSs are constructor-admitted at both budgets. Sionna2.2.0 uses CPU FP32, one 5G-NR LDPC block and RV0: filler shortening, first-$2Z$ and tail puncturing, no repetition. Decoder settings are boxplus-phi/check, sum/variable, flooding, 20 iterations, LLR clipping 20, hard output and information bits returned, PCM pruning on, HARQ and early stopping off. Gray-mapped QAM has nominal $E_s = 2$ without per-frame renormalization. Public PCG64 scrambling follows rate matching; RX reverses its LLR signs. Its seed is derived from the first 16 SHA256 bytes, little-endian, of the canonical protocol/"scramble"/session/counter tuple. Exact mask and mapping definitions are in `ep_phy.py`; interleaver hashes and constructor metadata are in `phy_layouts.csv`.

N	q	r	k	n	BG	Z	Filler	Punctured
512	2	1/2	444	888	bg2	56	116	1908
512	2	2/3	592	888	bg2	72	128	2728
512	2	3/4	666	888	bg1	32	38	1250
512	2	5/6	740	888	bg1	36	52	1508
512	4	1/2	888	1776	bg2	96	72	3144
512	4	2/3	1184	1776	bg2	120	16	4448
512	4	3/4	1332	1776	bg1	64	76	2500
512	4	5/6	1480	1776	bg1	72	104	3016
512	6	1/2	1332	2664	bg2	144	108	4716
512	6	2/3	1776	2664	bg2	192	144	7176
512	6	3/4	1998	2664	bg1	96	114	3750
512	6	5/6	2220	2664	bg1	104	68	4340
1024	2	1/2	956	1912	bg2	96	4	3076
1024	2	2/3	1274	1912	bg2	128	6	4738
1024	2	3/4	1434	1912	bg1	72	150	2834
1024	2	5/6	1593	1912	bg1	80	167	3361
1024	4	1/2	1912	3824	bg2	192	8	6152
1024	4	2/3	2549	3824	bg2	256	11	9477
1024	4	3/4	2868	3824	bg1	144	300	5668
1024	4	5/6	3186	3824	bg1	160	334	6722
1024	6	1/2	2868	5736	bg2	288	12	9228
1024	6	2/3	3824	5736	bg2	384	16	14216
1024	6	3/4	4302	5736	bg1	208	274	8134
1024	6	5/6	4780	5736	bg1	224	148	9348

Profile CRC/catalogue, body CRC, invalid length, known padding or arithmetic canonical failures produce fixed gray RGB 0.5. A valid paid-NULL also produces gray, while preserving its full transmitted header/body and statistical denominator. The current N1024 directory retains all 3378 original profiles and reuses five existing NULL IDs; seven missing MCS-specific NULLs occupy free IDs 3390–3396. Each carries payload [0,0]. The 33-check CPU regression performed 21 real header decodes and 20 body decodes, including every MCS no-fit path, exact capacity, one-bit excess,

legal fallbacks, normal-path bit/wave equality and malformed packets. No no-fit event occurred in the reported quality evaluations.

S1.4 Candidate pools and calibration

The requested MCS grid is $q \in \{2, 4, 6\}$, $r \in \{1/2, 2/3, 3/4, 5/6\}$. The `candidate_pool.csv` records the NeST-family pools, all N512 families and the N1024 7-dB candidates, including target, MCS, capacity, admitted status, screening/final qualification, winner, objective, tie rule and registration source. N512 screens 396 candidates: 25 Quarter targets×12, seven WHOLE targets×12 and Ada's maximum 256×12. Each candidate uses 100 calibration sources and one channel event. Per family/SNR, the top three plus the predefined maximum-endpoint $q2, r1/2$ anchor enter cal1000 with three events per source. Selection maximizes equal-source DINOv2-L cosine including gray failures; ties use lexical candidate ID.

The NeST-family N1024 policies at 4/10/19 dB use the actual inherited finalists plus the prescribed full-$m = 10$ candidates; ties prefer legacy candidates and then lexical ID. At 7 dB the five MCSs are $(2, 2/3), (2, 3/4), (2, 5/6), (4, 1/2), (4, 2/3)$, on 100 calibration sources with one event; ties follow declaration order. Token retains the selected family target/MCS and enlarges fallback endpoints only. Equal-bit controls instead share ceiling $(10, 0)$; the matched-granularity control shares $(8, 126)$. These are separate comparisons.

For Ada at $N = 1024$, the two (q, r) candidates at 4/10/19 dB are $\{(2, 1/2), (2, 2/3)\}$, $\{(4, 2/3), (4, 3/4)\}$ and $\{(4, 2/3), (6, 5/6)\}$, respectively. Selection on cal1000 with three events per source freezes $(2, 2/3)$, $(4, 2/3)$ and $(6, 5/6)$; it maximizes equal-source DINOv2-L cosine with lexical candidate-ID ties. The six scores and frozen policies are copied unchanged to `inputs/reproduction/ada_n1024_calibration/` as `candidate_summary.csv` and `frozen_policies.json`.

N	SNR	NeST target	WHOLE target	Modulation	Rate	Source cap.
1024	4	(7,25)	(7,0)	4-QAM	2/3	1245
1024	7	(10,0)	(10,0)	16-QAM	1/2	1883
1024	10	(8,126)	(9,0)	16-QAM	2/3	2520
1024	19	(10,0)	(10,0)	64-QAM	5/6	4751
512	4	(5,27)	(6,0)	4-QAM	2/3	563
512	10	(7,0)	(7,0)	16-QAM	2/3	1155
512	19	(8,84)	(8,0)	64-QAM	5/6	2191

Source capacity excludes the body length/CRC but is not total air-interface efficiency. Ada can saturate its full 256-token representation and leave capacity unused. Budgets have independently calibrated pools. DINOv2 is the policy objective; DISTS/DreamSim are source-reference evaluation measures, not human judgments. DISTS uses the official revision `1267d8cb626c98706db3697422701c56a85ebf2e`, learned alpha/beta and VGG16 weights, FP32 evaluation on native 256 RGB. DreamSim/CLIP use 224 bicubic-antialias; metric identities pin exact weights and dependencies. CLIP2.10 features are not mixed with 2.11 caches.

S1.5 Reference vectors and verification

Six deterministic CPU probability cases and six independently terminated prefixes verify positive integer frequencies, total frequency, first-index ties, non-mutating finalization, independent decoding, exact 30-bit extension and rejection of noncanonical appended bits. Run from the repository root:

```
python experiments/nestcom-v9-completion-20261101/verify_cdf_ac.py
-repo .  -output /tmp/nest-cdf-check
```

The output directory must be fresh. Two real-model vectors below record complete TX/RX scale-CDF hashes, exact bits/tokens, canonical checks, model/numerical identity and source provenance. RX did not consume TX probability tables, source tokens or hidden source states. Comparisons were made only after independent reception completed.

Vector	(m, K)	Tokens	Source bits	CDF scales	Exact TX/RX
Whole	(4, 0)	30	318	4	yes
Boundary	(6, 34)	125	1241	7	yes

The whole reference uses one explicitly counted receiver diagnostic without a new PHY event; the boundary reference reuses a completed Kodak input. Integer vectors replay without weights. Independent model/CDF regeneration requires the pinned assets and numerical environment; reference hashes characterize that tested route.

Common500 uses its original 10 000×500 source-bootstrap bank; controls use a separately ordered 10 000×100 bank. Kodak uses the existing 10 000×24 bank. Each source first averages its three events, then receives equal weight; paired differences precede resampling. Full source IDs, pixel hashes and native receiver identities control reuse. No simultaneous multiplicity-adjusted interval claim is made.

S2 Complete quality results

ARPC + LDPC applies our common digital frame to the ARPC progressive source codec. Its transmitted caption and other required fields are charged to the frame budget. Fig. S1 reports this additional comparison alongside the methods in the main Fig. 2. This section gives the complete quality results; S7 reports processing costs for the same adaptation.

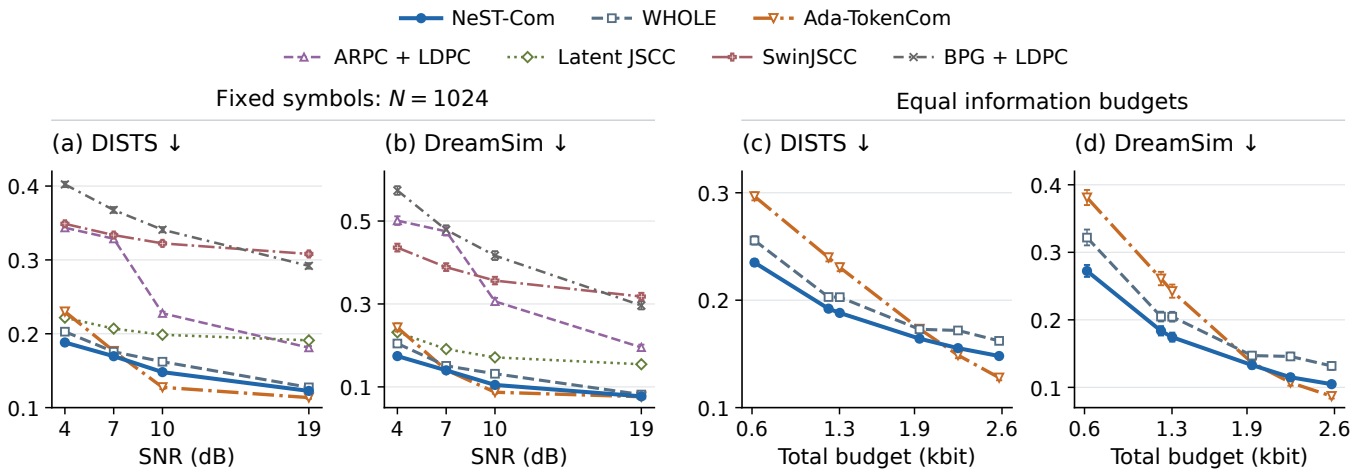


Figure S1: Complete comparison, including the digital-link adaptation of the ARPC source codec. Panels (a,b) retain all seven methods at $N = 1024$ and 4/7/10/19 dB; panels (c,d) retain all six equal-information budgets for NeST-Com, WHOLE and Ada. The 92 points and error bars are equal-source means and pointwise 95% source-bootstrap intervals after within-source averaging. Ada is the predicted-class, fixed-frame adaptation.

S2.1 Five-metric source means

All means retain failures. Pointwise intervals and direct paired intervals are provided in the full-precision CSVs; the compact table reports means.

N	SNR	Method	PSNR↑	LPIPS↓	DreamSim↓	CLIP-I↑	DISTS↓
512	4	Ada	11.318	0.5542	0.3847	0.7892	0.2996
512	4	WHOLE	15.107	0.3951	0.3247	0.7469	0.2580
512	4	NeST	15.611	0.3565	0.2740	0.7691	0.2363
512	4	Quarter	15.499	0.3652	0.2819	0.7657	0.2406
512	10	Ada	13.165	0.4110	0.2582	0.8257	0.2367
512	10	WHOLE	16.821	0.2881	0.2049	0.8074	0.2030

N	SNR	Method	PSNR↑	LPIPS↓	DreamSim↓	CLIP-I↑	DISTS↓
512	10	NeST	17.413	0.2650	0.1837	0.8211	0.1922
512	10	Quarter	17.220	0.2726	0.1922	0.8150	0.1958
512	19	Ada	17.690	0.2146	0.1078	0.8723	0.1496
512	19	WHOLE	18.638	0.2175	0.1458	0.8484	0.1718
512	19	NeST	19.632	0.1832	0.1150	0.8689	0.1554
512	19	Quarter	19.430	0.1901	0.1217	0.8640	0.1587
1024	4	ARPC	14.184	0.5124	0.5010	0.6442	0.3436
1024	4	Ada	13.332	0.3952	0.2431	0.8280	0.2300
1024	4	BPG	20.743	0.7255	0.5737	0.5859	0.4022
1024	4	Latent	19.658	0.2188	0.2317	0.7408	0.2218
1024	4	Swin	21.671	0.5297	0.4359	0.6267	0.3488
1024	4	WHOLE	16.832	0.2878	0.2046	0.8077	0.2028
1024	4	NeST	17.654	0.2549	0.1743	0.8271	0.1882
1024	4	Quarter	17.448	0.2633	0.1839	0.8206	0.1916
1024	7	ARPC	14.759	0.4809	0.4751	0.6508	0.3287
1024	7	Ada	15.905	0.2698	0.1408	0.8592	0.1766
1024	7	BPG	21.717	0.6500	0.4788	0.6150	0.3676
1024	7	Latent	20.369	0.1893	0.1912	0.7717	0.2070
1024	7	Swin	22.475	0.4933	0.3887	0.6505	0.3336
1024	7	WHOLE	18.528	0.2228	0.1507	0.8460	0.1756
1024	7	NeST	18.981	0.2085	0.1399	0.8539	0.1696
1024	7	Quarter	18.850	0.2114	0.1420	0.8507	0.1699
1024	10	ARPC	16.471	0.3413	0.3062	0.7134	0.2278
1024	10	Ada	19.559	0.1674	0.0871	0.8841	0.1275
1024	10	BPG	22.543	0.5894	0.4165	0.6350	0.3409
1024	10	Latent	20.803	0.1733	0.1712	0.7892	0.1986
1024	10	Swin	23.026	0.4687	0.3564	0.6691	0.3224
1024	10	WHOLE	19.313	0.1987	0.1317	0.8577	0.1621
1024	10	NeST	20.194	0.1658	0.1048	0.8769	0.1481
1024	10	Quarter	20.014	0.1715	0.1077	0.8731	0.1504
1024	19	ARPC	18.199	0.2503	0.1952	0.7880	0.1812
1024	19	Ada	20.938	0.1376	0.0762	0.8925	0.1136
1024	19	BPG	24.230	0.4584	0.2964	0.6843	0.2920
1024	19	Latent	21.245	0.1585	0.1546	0.8041	0.1911
1024	19	Swin	23.698	0.4400	0.3179	0.6966	0.3081
1024	19	WHOLE	21.844	0.1234	0.0815	0.8969	0.1274
1024	19	NeST	22.334	0.1151	0.0772	0.9015	0.1226
1024	19	Quarter	22.164	0.1182	0.0785	0.9005	0.1243

S2.2 Direct paired differences

NeST minus the stated reference. Positive PSNR/CLIP-I and negative distance differences favor NeST. A confidence interval spanning zero does not establish a direction.

N	SNR	Reference	Metric	Difference [95% CI]
512	4	Quarter	PSNR	+0.11210 [0.06798, 0.16080]
512	4	Quarter	LPIPS	-0.00863 [-0.01104, -0.00621]
512	4	Quarter	DreamSim	-0.00782 [-0.01132, -0.00447]
512	4	Quarter	CLIP-I	+0.00345 [-0.00043, 0.00738]
512	4	Quarter	DISTS	-0.00433 [-0.00627, -0.00247]
512	10	Quarter	PSNR	+0.19304 [0.16224, 0.22489]
512	10	Quarter	LPIPS	-0.00759 [-0.00914, -0.00622]
512	10	Quarter	DreamSim	-0.00842 [-0.01061, -0.00630]
512	10	Quarter	CLIP-I	+0.00611 [0.00330, 0.00898]
512	10	Quarter	DISTS	-0.00367 [-0.00476, -0.00270]

N	SNR	Reference	Metric	Difference [95% CI]
512	19	Quarter	PSNR	+0.20191 [0.17590, 0.22869]
512	19	Quarter	LPIPS	-0.00681 [-0.00771, -0.00595]
512	19	Quarter	DreamSim	-0.00669 [-0.00828, -0.00521]
512	19	Quarter	CLIP-I	+0.00489 [0.00239, 0.00738]
512	19	Quarter	DISTS	-0.00326 [-0.00391, -0.00263]
1024	4	Quarter	PSNR	+0.20601 [0.17556, 0.23714]
1024	4	Quarter	LPIPS	-0.00831 [-0.00935, -0.00728]
1024	4	Quarter	DreamSim	-0.00960 [-0.01196, -0.00732]
1024	4	Quarter	CLIP-I	+0.00648 [0.00334, 0.00973]
1024	4	Quarter	DISTS	-0.00343 [-0.00420, -0.00268]
1024	7	Quarter	PSNR	+0.13138 [0.07917, 0.18093]
1024	7	Quarter	LPIPS	-0.00292 [-0.00689, 0.00135]
1024	7	Quarter	DreamSim	-0.00210 [-0.00668, 0.00274]
1024	7	Quarter	CLIP-I	+0.00318 [-0.00002, 0.00639]
1024	7	Quarter	DISTS	-0.00024 [-0.00368, 0.00344]
1024	10	Quarter	PSNR	+0.18002 [0.15807, 0.20281]
1024	10	Quarter	LPIPS	-0.00567 [-0.00635, -0.00502]
1024	10	Quarter	DreamSim	-0.00292 [-0.00379, -0.00204]
1024	10	Quarter	CLIP-I	+0.00379 [0.00178, 0.00581]
1024	10	Quarter	DISTS	-0.00235 [-0.00286, -0.00185]
1024	19	Quarter	PSNR	+0.17033 [0.15600, 0.18558]
1024	19	Quarter	LPIPS	-0.00304 [-0.00332, -0.00277]
1024	19	Quarter	DreamSim	-0.00127 [-0.00164, -0.00090]
1024	19	Quarter	CLIP-I	+0.00100 [-0.00007, 0.00209]
1024	19	Quarter	DISTS	-0.00166 [-0.00194, -0.00138]
512	4	WHOLE	PSNR	+0.50427 [0.43976, 0.57335]
512	4	WHOLE	LPIPS	-0.03858 [-0.04275, -0.03445]
512	4	WHOLE	DreamSim	-0.05062 [-0.05665, -0.04454]
512	4	WHOLE	CLIP-I	+0.02219 [0.01714, 0.02740]
512	4	WHOLE	DISTS	-0.02171 [-0.02449, -0.01904]
512	10	WHOLE	PSNR	+0.59161 [0.54369, 0.64265]
512	10	WHOLE	LPIPS	-0.02307 [-0.02452, -0.02161]
512	10	WHOLE	DreamSim	-0.02112 [-0.02363, -0.01865]
512	10	WHOLE	CLIP-I	+0.01365 [0.01009, 0.01718]
512	10	WHOLE	DISTS	-0.01082 [-0.01180, -0.00986]
512	19	WHOLE	PSNR	+0.99391 [0.93976, 1.04841]
512	19	WHOLE	LPIPS	-0.03430 [-0.03577, -0.03284]
512	19	WHOLE	DreamSim	-0.03081 [-0.03328, -0.02843]
512	19	WHOLE	CLIP-I	+0.02054 [0.01696, 0.02425]
512	19	WHOLE	DISTS	-0.01644 [-0.01747, -0.01542]
1024	4	WHOLE	PSNR	+0.82204 [0.76554, 0.88112]
1024	4	WHOLE	LPIPS	-0.03289 [-0.03452, -0.03128]
1024	4	WHOLE	DreamSim	-0.03033 [-0.03335, -0.02731]
1024	4	WHOLE	CLIP-I	+0.01942 [0.01560, 0.02316]
1024	4	WHOLE	DISTS	-0.01463 [-0.01571, -0.01357]
1024	7	WHOLE	PSNR	+0.45308 [0.39299, 0.51258]
1024	7	WHOLE	LPIPS	-0.01432 [-0.01880, -0.00967]
1024	7	WHOLE	DreamSim	-0.01084 [-0.01567, -0.00566]
1024	7	WHOLE	CLIP-I	+0.00793 [0.00439, 0.01153]
1024	7	WHOLE	DISTS	-0.00603 [-0.00966, -0.00207]
1024	10	WHOLE	PSNR	+0.88117 [0.80754, 0.95550]
1024	10	WHOLE	LPIPS	-0.03293 [-0.03561, -0.03021]
1024	10	WHOLE	DreamSim	-0.02697 [-0.02990, -0.02408]
1024	10	WHOLE	CLIP-I	+0.01919 [0.01552, 0.02307]
1024	10	WHOLE	DISTS	-0.01406 [-0.01531, -0.01280]
1024	19	WHOLE	PSNR	+0.48975 [0.45749, 0.52298]
1024	19	WHOLE	LPIPS	-0.00824 [-0.00868, -0.00782]

N	SNR	Reference	Metric	Difference [95% CI]
1024	19	WHOLE	DreamSim	-0.00431 [-0.00508, -0.00359]
1024	19	WHOLE	CLIP-I	+0.00463 [0.00302, 0.00624]
1024	19	WHOLE	DISTS	-0.00482 [-0.00532, -0.00432]
512	4	Ada	PSNR	+4.29250 [4.09816, 4.49208]
512	4	Ada	LPIPS	-0.19761 [-0.20524, -0.19011]
512	4	Ada	DreamSim	-0.11068 [-0.12210, -0.09964]
512	4	Ada	CLIP-I	-0.02007 [-0.02843, -0.01175]
512	4	Ada	DISTS	-0.06332 [-0.06766, -0.05893]
512	10	Ada	PSNR	+4.24824 [4.07079, 4.42766]
512	10	Ada	LPIPS	-0.14597 [-0.15149, -0.14058]
512	10	Ada	DreamSim	-0.07443 [-0.08290, -0.06583]
512	10	Ada	CLIP-I	-0.00466 [-0.01139, 0.00187]
512	10	Ada	DISTS	-0.04455 [-0.04757, -0.04151]
512	19	Ada	PSNR	+1.94166 [1.77815, 2.11000]
512	19	Ada	LPIPS	-0.03133 [-0.03469, -0.02813]
512	19	Ada	DreamSim	+0.00723 [0.00374, 0.01056]
512	19	Ada	CLIP-I	-0.00339 [-0.00816, 0.00132]
512	19	Ada	DISTS	+0.00581 [0.00364, 0.00788]
1024	4	Ada	PSNR	+4.32222 [4.12839, 4.52292]
1024	4	Ada	LPIPS	-0.14026 [-0.14581, -0.13482]
1024	4	Ada	DreamSim	-0.06883 [-0.07668, -0.06109]
1024	4	Ada	CLIP-I	-0.00089 [-0.00713, 0.00524]
1024	4	Ada	DISTS	-0.04183 [-0.04498, -0.03863]
1024	7	Ada	PSNR	+3.07575 [2.88099, 3.27370]
1024	7	Ada	LPIPS	-0.06131 [-0.06696, -0.05571]
1024	7	Ada	DreamSim	-0.00093 [-0.00770, 0.00573]
1024	7	Ada	CLIP-I	-0.00527 [-0.01070, -0.00003]
1024	7	Ada	DISTS	-0.00701 [-0.01123, -0.00274]
1024	10	Ada	PSNR	+0.63508 [0.52112, 0.75347]
1024	10	Ada	LPIPS	-0.00156 [-0.00407, 0.00081]
1024	10	Ada	DreamSim	+0.01766 [0.01532, 0.01994]
1024	10	Ada	CLIP-I	-0.00721 [-0.01140, -0.00307]
1024	10	Ada	DISTS	+0.02055 [0.01885, 0.02218]
1024	19	Ada	PSNR	+1.39662 [1.33666, 1.45531]
1024	19	Ada	LPIPS	-0.02242 [-0.02342, -0.02140]
1024	19	Ada	DreamSim	+0.00095 [-0.00086, 0.00278]
1024	19	Ada	CLIP-I	+0.00903 [0.00521, 0.01290]
1024	19	Ada	DISTS	+0.00904 [0.00762, 0.01050]

S2.3 Failures and denominator

The following nonzero gray-image counts are verified from native pixel hashes and receiver statuses. Other cells have zero observed gray outputs. Main systems use their prescribed independent noise streams; the matched-granularity control uses shared noise.

N	SNR	Method	Gray count	Denominator
512	4	Ada	5	1500
1024	7	Ada	7	1500
1024	7	BPG	11	1500
512	4	WHOLE	7	1500
1024	7	WHOLE	7	1500
512	4	NeST	4	1500
1024	7	NeST	13	1500
512	4	Quarter	4	1500
512	10	Quarter	1	1500

N	SNR	Method	Gray count	Denominator
1024	7	Quarter	6	1500

S2.4 Per-source improvement distributions

Gain is target minus reference for PSNR/CLIP-I and reference minus target for distances. Better/tie/worse uses numerical tolerance 10^{-10}; this is not a perceptual visibility threshold. Quantiles use the linear rule. Rows retain all sources and failures. N1024/10 Token versus WHOLE improves DISTS for 83.4%, DreamSim for 83.0%, and both for 72.2%; each distance worsens on 16.2%. These are descriptive proportions, alongside the direct paired mean intervals above.

N/SNR	Reference	Metric	Better/tie/worse	P05	Median	P95
512/4	WHOLE	PSNR	362/42/96	-0.2496	+0.2694	+1.7441
512/4	WHOLE	LPIPS	381/42/77	-0.0085	+0.0196	+0.1291
512/4	WHOLE	DreamSim	361/42/97	-0.0246	+0.0250	+0.1953
512/4	WHOLE	CLIP-I	299/42/159	-0.0573	+0.0094	+0.1290
512/4	WHOLE	DISTS	375/42/83	-0.0082	+0.0138	+0.0679
512/4	Quarter	PSNR	270/60/170	-0.4538	+0.0249	+0.8224
512/4	Quarter	LPIPS	313/60/127	-0.0134	+0.0053	+0.0389
512/4	Quarter	DreamSim	265/60/175	-0.0428	+0.0020	+0.0681
512/4	Quarter	CLIP-I	236/60/204	-0.0663	+0.0000	+0.0707
512/4	Quarter	DISTS	294/60/146	-0.0129	+0.0025	+0.0247
512/4	Ada	PSNR	496/0/4	+1.0635	+4.1103	+8.1459
512/4	Ada	LPIPS	497/0/3	+0.0657	+0.1931	+0.3573
512/4	Ada	DreamSim	413/0/87	-0.0803	+0.1046	+0.3274
512/4	Ada	CLIP-I	217/0/283	-0.1886	-0.0132	+0.1280
512/4	Ada	DISTS	466/0/34	-0.0052	+0.0603	+0.1516
512/10	WHOLE	PSNR	474/2/24	+0.0000	+0.4505	+1.7163
512/10	WHOLE	LPIPS	472/2/26	-0.0004	+0.0212	+0.0540
512/10	WHOLE	DreamSim	389/2/109	-0.0178	+0.0179	+0.0798
512/10	WHOLE	CLIP-I	329/2/169	-0.0498	+0.0093	+0.0868
512/10	WHOLE	DISTS	423/2/75	-0.0055	+0.0103	+0.0296
512/10	Quarter	PSNR	345/32/123	-0.2317	+0.1260	+0.8049
512/10	Quarter	LPIPS	345/32/123	-0.0096	+0.0057	+0.0293
512/10	Quarter	DreamSim	298/32/170	-0.0239	+0.0053	+0.0439
512/10	Quarter	CLIP-I	268/32/200	-0.0431	+0.0021	+0.0614
512/10	Quarter	DISTS	304/32/164	-0.0094	+0.0022	+0.0201
512/10	Ada	PSNR	500/0/0	+1.4835	+3.9296	+8.1875
512/10	Ada	LPIPS	499/0/1	+0.0523	+0.1409	+0.2573
512/10	Ada	DreamSim	397/0/103	-0.0560	+0.0608	+0.2530
512/10	Ada	CLIP-I	256/0/244	-0.1487	+0.0013	+0.1180
512/10	Ada	DISTS	458/0/42	-0.0088	+0.0417	+0.1011
512/19	WHOLE	PSNR	491/2/7	+0.2019	+0.8817	+2.2591
512/19	WHOLE	LPIPS	493/2/5	+0.0072	+0.0336	+0.0628
512/19	WHOLE	DreamSim	471/2/27	-0.0004	+0.0254	+0.0841
512/19	WHOLE	CLIP-I	357/2/141	-0.0329	+0.0132	+0.0995
512/19	WHOLE	DISTS	462/2/36	-0.0006	+0.0161	+0.0379
512/19	Quarter	PSNR	379/23/98	-0.1316	+0.1230	+0.8131
512/19	Quarter	LPIPS	370/23/107	-0.0061	+0.0053	+0.0257
512/19	Quarter	DreamSim	322/23/155	-0.0139	+0.0035	+0.0383
512/19	Quarter	CLIP-I	276/23/201	-0.0332	+0.0022	+0.0520
512/19	Quarter	DISTS	331/23/146	-0.0083	+0.0024	+0.0160
512/19	Ada	PSNR	444/0/56	-0.5606	+1.6364	+5.6562
512/19	Ada	LPIPS	399/0/101	-0.0208	+0.0274	+0.0977
512/19	Ada	DreamSim	165/0/335	-0.0633	-0.0082	+0.0499
512/19	Ada	CLIP-I	237/0/263	-0.0912	-0.0018	+0.0784
512/19	Ada	DISTS	173/0/327	-0.0411	-0.0071	+0.0377

N/SNR	Reference	Metric	Better/tie/worse	P05	Median	P95
1024/4	WHOLE	PSNR	488/0/12	+0.0976	+0.6558	+2.1413
1024/4	WHOLE	LPIPS	492/0/8	+0.0055	+0.0308	+0.0658
1024/4	WHOLE	DreamSim	432/0/68	-0.0147	+0.0242	+0.0975
1024/4	WHOLE	CLIP-I	345/0/155	-0.0430	+0.0158	+0.0931
1024/4	WHOLE	DISTS	447/0/53	-0.0039	+0.0136	+0.0362
1024/4	Quarter	PSNR	374/0/126	-0.2168	+0.1524	+0.8926
1024/4	Quarter	LPIPS	366/0/134	-0.0073	+0.0064	+0.0301
1024/4	Quarter	DreamSim	327/11/162	-0.0249	+0.0054	+0.0559
1024/4	Quarter	CLIP-I	272/2/226	-0.0457	+0.0012	+0.0613
1024/4	Quarter	DISTS	318/10/172	-0.0094	+0.0024	+0.0176
1024/4	Ada	PSNR	497/0/3	+1.3662	+3.9578	+8.5143
1024/4	Ada	LPIPS	497/0/3	+0.0453	+0.1362	+0.2457
1024/4	Ada	DreamSim	403/0/97	-0.0471	+0.0541	+0.2385
1024/4	Ada	CLIP-I	268/0/232	-0.1292	+0.0048	+0.1108
1024/4	Ada	DISTS	446/0/54	-0.0091	+0.0386	+0.1069
1024/7	WHOLE	PSNR	461/0/39	-0.0594	+0.3809	+1.4071
1024/7	WHOLE	LPIPS	469/0/31	-0.0019	+0.0158	+0.0401
1024/7	WHOLE	DreamSim	398/0/102	-0.0150	+0.0109	+0.0540
1024/7	WHOLE	CLIP-I	306/0/194	-0.0568	+0.0069	+0.0705
1024/7	WHOLE	DISTS	408/0/92	-0.0085	+0.0077	+0.0265
1024/7	Quarter	PSNR	359/25/116	-0.1978	+0.1076	+0.7268
1024/7	Quarter	LPIPS	365/25/110	-0.0075	+0.0045	+0.0237
1024/7	Quarter	DreamSim	309/25/166	-0.0183	+0.0030	+0.0363
1024/7	Quarter	CLIP-I	254/25/221	-0.0532	+0.0003	+0.0574
1024/7	Quarter	DISTS	327/25/148	-0.0101	+0.0023	+0.0177
1024/7	Ada	PSNR	471/0/29	-0.1458	+2.7760	+7.0974
1024/7	Ada	LPIPS	454/0/46	-0.0131	+0.0602	+0.1545
1024/7	Ada	DreamSim	233/0/267	-0.0821	-0.0026	+0.1231
1024/7	Ada	CLIP-I	236/0/264	-0.1079	-0.0059	+0.0964
1024/7	Ada	DISTS	293/0/207	-0.0371	+0.0063	+0.0661
1024/10	WHOLE	PSNR	438/0/62	-0.0182	+0.7597	+2.3170
1024/10	WHOLE	LPIPS	439/0/61	-0.0013	+0.0365	+0.0829
1024/10	WHOLE	DreamSim	415/4/81	-0.0032	+0.0148	+0.0898
1024/10	WHOLE	CLIP-I	344/0/156	-0.0288	+0.0064	+0.0979
1024/10	WHOLE	DISTS	417/2/81	-0.0024	+0.0113	+0.0400
1024/10	Quarter	PSNR	391/0/109	-0.0622	+0.1011	+0.6810
1024/10	Quarter	LPIPS	396/0/104	-0.0036	+0.0042	+0.0194
1024/10	Quarter	DreamSim	320/5/175	-0.0116	+0.0015	+0.0215
1024/10	Quarter	CLIP-I	284/0/216	-0.0278	+0.0012	+0.0439
1024/10	Quarter	DISTS	329/5/166	-0.0056	+0.0016	+0.0132
1024/10	Ada	PSNR	345/0/155	-1.2376	+0.4750	+3.0169
1024/10	Ada	LPIPS	226/0/274	-0.0349	-0.0026	+0.0542
1024/10	Ada	DreamSim	100/0/400	-0.0599	-0.0149	+0.0165
1024/10	Ada	CLIP-I	205/0/295	-0.0849	-0.0074	+0.0636
1024/10	Ada	DISTS	62/0/438	-0.0496	-0.0220	+0.0093
1024/19	WHOLE	PSNR	496/0/4	+0.0867	+0.3851	+1.2180
1024/19	WHOLE	LPIPS	492/0/8	+0.0012	+0.0077	+0.0168
1024/19	WHOLE	DreamSim	361/3/136	-0.0047	+0.0030	+0.0162
1024/19	WHOLE	CLIP-I	299/1/200	-0.0171	+0.0017	+0.0366
1024/19	WHOLE	DISTS	413/1/86	-0.0026	+0.0043	+0.0144
1024/19	Quarter	PSNR	472/0/28	-0.0000	+0.1358	+0.4535
1024/19	Quarter	LPIPS	430/0/70	-0.0009	+0.0025	+0.0085
1024/19	Quarter	DreamSim	308/8/184	-0.0037	+0.0006	+0.0080
1024/19	Quarter	CLIP-I	258/3/239	-0.0144	+0.0000	+0.0188
1024/19	Quarter	DISTS	350/4/146	-0.0033	+0.0013	+0.0076
1024/19	Ada	PSNR	493/0/7	+0.4733	+1.3082	+2.5644
1024/19	Ada	LPIPS	492/0/8	+0.0046	+0.0215	+0.0428

N/SNR	Reference	Metric	Better/tie/worse	P05	Median	P95
1024/19	Ada	DreamSim	232/0/268	-0.0333	-0.0011	+0.0327
1024/19	Ada	CLIP-I	291/0/209	-0.0578	+0.0072	+0.0861
1024/19	Ada	DISTS	144/0/356	-0.0378	-0.0081	+0.0181

Bits	Reference	Metric	Better/tie/worse	P05	Median	P95
620	WHOLE	PSNR	361/42/97	-0.2496	+0.2694	+1.7003
620	WHOLE	LPIPS	381/42/77	-0.0080	+0.0192	+0.1214
620	WHOLE	DreamSim	360/42/98	-0.0222	+0.0243	+0.1910
620	WHOLE	CLIP-I	297/42/161	-0.0605	+0.0092	+0.1262
620	WHOLE	DISTS	374/42/84	-0.0078	+0.0134	+0.0662
620	Ada	PSNR	499/0/1	+1.4708	+4.0982	+7.5756
620	Ada	LPIPS	497/0/3	+0.0701	+0.1923	+0.3280
620	Ada	DreamSim	418/0/82	-0.0760	+0.1040	+0.3216
620	Ada	CLIP-I	219/0/281	-0.1831	-0.0109	+0.1297
620	Ada	DISTS	474/0/26	-0.0005	+0.0580	+0.1265
1212	WHOLE	PSNR	475/0/25	+0.0003	+0.4509	+1.7163
1212	WHOLE	LPIPS	473/0/27	-0.0006	+0.0212	+0.0540
1212	WHOLE	DreamSim	391/0/109	-0.0178	+0.0179	+0.0798
1212	WHOLE	CLIP-I	330/0/170	-0.0498	+0.0098	+0.0868
1212	WHOLE	DISTS	424/0/76	-0.0055	+0.0104	+0.0299
1212	Ada	PSNR	499/0/1	+1.7129	+3.9729	+8.1332
1212	Ada	LPIPS	498/0/2	+0.0560	+0.1460	+0.2381
1212	Ada	DreamSim	419/0/81	-0.0523	+0.0655	+0.2320
1212	Ada	CLIP-I	275/0/225	-0.1235	+0.0084	+0.1062
1212	Ada	DISTS	464/0/36	-0.0055	+0.0461	+0.1114
1302	WHOLE	PSNR	488/1/11	+0.1039	+0.6558	+2.1413
1302	WHOLE	LPIPS	490/1/9	+0.0056	+0.0307	+0.0658
1302	WHOLE	DreamSim	431/1/68	-0.0147	+0.0242	+0.0975
1302	WHOLE	CLIP-I	346/1/153	-0.0430	+0.0164	+0.0931
1302	WHOLE	DISTS	446/1/53	-0.0039	+0.0136	+0.0362
1302	Ada	PSNR	499/0/1	+1.6604	+3.9410	+8.1762
1302	Ada	LPIPS	499/0/1	+0.0558	+0.1393	+0.2309
1302	Ada	DreamSim	408/0/92	-0.0406	+0.0580	+0.2283
1302	Ada	CLIP-I	264/0/236	-0.1289	+0.0039	+0.0916
1302	Ada	DISTS	464/0/36	-0.0085	+0.0433	+0.1030
1940	WHOLE	PSNR	473/0/27	-0.0050	+0.3948	+1.3201
1940	WHOLE	LPIPS	481/0/19	+0.0007	+0.0159	+0.0370
1940	WHOLE	DreamSim	407/0/93	-0.0105	+0.0111	+0.0486
1940	WHOLE	CLIP-I	312/0/188	-0.0421	+0.0072	+0.0631
1940	WHOLE	DISTS	419/0/81	-0.0056	+0.0078	+0.0254
1940	Ada	PSNR	482/0/18	+0.1806	+2.7900	+6.5027
1940	Ada	LPIPS	471/0/29	-0.0009	+0.0596	+0.1437
1940	Ada	DreamSim	248/0/252	-0.0672	-0.0008	+0.0839
1940	Ada	CLIP-I	237/0/263	-0.0991	-0.0025	+0.0718
1940	Ada	DISTS	305/0/195	-0.0322	+0.0070	+0.0561
2248	WHOLE	PSNR	491/2/7	+0.2019	+0.8817	+2.2593
2248	WHOLE	LPIPS	493/2/5	+0.0072	+0.0336	+0.0628
2248	WHOLE	DreamSim	472/2/26	-0.0003	+0.0254	+0.0841
2248	WHOLE	CLIP-I	357/2/141	-0.0329	+0.0132	+0.0995
2248	WHOLE	DISTS	463/2/35	-0.0006	+0.0161	+0.0379
2248	Ada	PSNR	448/0/52	-0.4519	+1.7167	+4.8464
2248	Ada	LPIPS	406/0/94	-0.0201	+0.0261	+0.0946
2248	Ada	DreamSim	165/0/335	-0.0586	-0.0098	+0.0476
2248	Ada	CLIP-I	235/0/265	-0.0913	-0.0035	+0.0760
2248	Ada	DISTS	166/0/334	-0.0415	-0.0074	+0.0301

Bits	Reference	Metric	Better/tie/worse	P05	Median	P95
2577	WHOLE	PSNR	432/16/52	-0.0182	+0.7685	+2.3170
2577	WHOLE	LPIPS	433/16/51	-0.0013	+0.0365	+0.0829
2577	WHOLE	DreamSim	409/16/75	-0.0032	+0.0148	+0.0898
2577	WHOLE	CLIP-I	337/16/147	-0.0288	+0.0064	+0.0979
2577	WHOLE	DISTS	413/16/71	-0.0023	+0.0113	+0.0400
2577	Ada	PSNR	353/0/147	-1.1817	+0.5351	+3.0938
2577	Ada	LPIPS	231/0/269	-0.0345	-0.0020	+0.0526
2577	Ada	DreamSim	99/0/401	-0.0594	-0.0156	+0.0176
2577	Ada	CLIP-I	228/0/272	-0.0846	-0.0044	+0.0591
2577	Ada	DISTS	63/0/437	-0.0482	-0.0211	+0.0094

S3 Matched controls and generation sensitivity

S3.1 Endpoint control

The four grids use control100, $N = 1024$, 10 dB, 16-QAM rate 2/3, a common maximum token position $(8, 126)$, and the actual shared noise vectors for repeats 6201–6203. Only legal termination points change. This WHOLE control has five permitted complete-scale endpoints; the main WHOLE system has a different target and must not be substituted. There are 1200 logical outcomes, 891 unique physical inputs, 591 newly executed physical events, and 300 exact physical reuses. Independent RX reconstruction uses 100 new and 197 reused receiver inputs.

Grid	Ends	Util. (%)	Length checks	DISTS	DreamSim
Whole	5	78.133	1.55	0.1650	0.1387
Quarter	20	95.194	4.51	0.1535	0.1143
Eighth	39	97.442	7.57	0.1516	0.1137
Token	352	99.848	125.96	0.1505	0.1115

The quality experiment reuses previously finalized streams, so its new source traversals and independent finalizations are zero. Candidate-set cardinality and descending comparisons must not be described as new network forwards. Fresh deployment computation and measured costs belong to S7. The legacy main-evaluation field named length_checks sometimes records the entire evaluated endpoint set; it is not substituted for the matched-control stopping count shown here.

Comparison	Metric	Difference [95% CI]
Quarter–Whole	PSNR	+0.65169 [0.50944, 0.80674]
Quarter–Whole	LPIPS	-0.02750 [-0.03334, -0.02195]
Quarter–Whole	DreamSim	-0.02447 [-0.03163, -0.01792]
Quarter–Whole	CLIP-I	+0.01678 [0.00815, 0.02602]
Quarter–Whole	DISTS	-0.01153 [-0.01410, -0.00905]
Eighth–Whole	PSNR	+0.75730 [0.59558, 0.93119]
Eighth–Whole	LPIPS	-0.03046 [-0.03672, -0.02443]
Eighth–Whole	DreamSim	-0.02505 [-0.03221, -0.01845]
Eighth–Whole	CLIP-I	+0.01834 [0.00979, 0.02735]
Eighth–Whole	DISTS	-0.01342 [-0.01623, -0.01065]
Token–Whole	PSNR	+0.82337 [0.65877, 1.00101]
Token–Whole	LPIPS	-0.03270 [-0.03923, -0.02649]
Token–Whole	DreamSim	-0.02725 [-0.03419, -0.02071]
Token–Whole	CLIP-I	+0.02339 [0.01484, 0.03271]
Token–Whole	DISTS	-0.01454 [-0.01743, -0.01167]
Eighth–Quarter	PSNR	+0.10561 [0.06774, 0.14656]
Eighth–Quarter	LPIPS	-0.00296 [-0.00408, -0.00194]
Eighth–Quarter	DreamSim	-0.00058 [-0.00201, 0.00082]
Eighth–Quarter	CLIP-I	+0.00156 [-0.00145, 0.00456]

Comparison	Metric	Difference [95% CI]
Eighth–Quarter	DISTS	-0.00189 [-0.00279, -0.00106]
Token–Quarter	PSNR	+0.17168 [0.12500, 0.22129]
Token–Quarter	LPIPS	-0.00520 [-0.00653, -0.00390]
Token–Quarter	DreamSim	-0.00278 [-0.00501, -0.00059]
Token–Quarter	CLIP-I	+0.00661 [0.00174, 0.01190]
Token–Quarter	DISTS	-0.00300 [-0.00411, -0.00195]
Token–Eighth	PSNR	+0.06607 [0.03357, 0.10063]
Token–Eighth	LPIPS	-0.00224 [-0.00318, -0.00131]
Token–Eighth	DreamSim	-0.00220 [-0.00422, -0.00014]
Token–Eighth	CLIP-I	+0.00505 [0.00022, 0.01051]
Token–Eighth	DISTS	-0.00112 [-0.00204, -0.00022]

S3.2 Ordered boundary-token interventions

A0 retains prior completion but omits the extra boundary-token correction and updated conditioning. B0 inserts the direct residual correction while keeping the A0 future-condition path. C0 applies both. All three arms use the same actual received prefix and retain failures. This is an ordered intervention on the full received boundary, not an additive decomposition of the tokens beyond Quarter. The complete set contains 1800 logical outcomes. It requires 394 new A0/B0 renders for 197 unique prefixes; two qualification C0 replays are pixel-identical to the official output. No new PHY events are generated.

SNR	Comparison	Metric	Difference [95% CI]
7	B0–A0	PSNR	+0.16706 [0.13852, 0.19873]
7	B0–A0	LPIPS	-0.00639 [-0.00718, -0.00558]
7	B0–A0	DreamSim	-0.00565 [-0.00774, -0.00376]
7	B0–A0	CLIP-I	-0.00061 [-0.00386, 0.00244]
7	B0–A0	DISTS	-0.00294 [-0.00383, -0.00205]
7	C0–B0	PSNR	+0.29197 [0.24095, 0.34673]
7	C0–B0	LPIPS	-0.01079 [-0.01266, -0.00895]
7	C0–B0	DreamSim	-0.00973 [-0.01344, -0.00626]
7	C0–B0	CLIP-I	+0.00860 [0.00275, 0.01511]
7	C0–B0	DISTS	-0.00456 [-0.00620, -0.00291]
7	C0–A0	PSNR	+0.45903 [0.39375, 0.52953]
7	C0–A0	LPIPS	-0.01717 [-0.01940, -0.01508]
7	C0–A0	DreamSim	-0.01538 [-0.01982, -0.01135]
7	C0–A0	CLIP-I	+0.00798 [0.00190, 0.01445]
7	C0–A0	DISTS	-0.00750 [-0.00927, -0.00576]
10	B0–A0	PSNR	+0.28681 [0.23964, 0.33531]
10	B0–A0	LPIPS	-0.01225 [-0.01448, -0.01007]
10	B0–A0	DreamSim	-0.01131 [-0.01468, -0.00828]
10	B0–A0	CLIP-I	+0.00190 [-0.00310, 0.00728]
10	B0–A0	DISTS	-0.00621 [-0.00769, -0.00481]
10	C0–B0	PSNR	+0.53657 [0.40447, 0.68156]
10	C0–B0	LPIPS	-0.02045 [-0.02521, -0.01597]
10	C0–B0	DreamSim	-0.01594 [-0.02051, -0.01161]
10	C0–B0	CLIP-I	+0.02149 [0.01389, 0.02963]
10	C0–B0	DISTS	-0.00833 [-0.01022, -0.00647]
10	C0–A0	PSNR	+0.82337 [0.65877, 1.00101]
10	C0–A0	LPIPS	-0.03270 [-0.03923, -0.02649]
10	C0–A0	DreamSim	-0.02725 [-0.03419, -0.02071]
10	C0–A0	CLIP-I	+0.02339 [0.01484, 0.03271]
10	C0–A0	DISTS	-0.01454 [-0.01743, -0.01167]

At $N = 1024$, 19 dB, 1482 of 1500 NeST events have a partial final-scale boundary and 18 transmit the full representation.

There is no finer scale to condition, hence B0 and C0 apply the identical final residual map. This structural check introduces no new renders, and historical Quarter A0/B0 means remain labelled Quarter.

S3.3 Ada sampled versus argmax completion

The same 100 source IDs, paid predicted classes, actual received prefixes, three original channel outcomes, models, CFG 4 and numerical route are fixed. Only unreceived-tail selection changes. The greedy context/KV is rebuilt and updated along its own generated path; conditional entropy CDFs do not use CFG. Completed full prefixes and gray outputs are reused. This is a receiver sensitivity control, not additional wireless frames. Table values are argmax minus sampling. At the three tested points, argmax increases both LPIPS and DreamSim. The main comparison retains the original sampling configuration.

N	SNR	Metric	Difference [paired 95% CI]
512	4	PSNR	-0.47494 [-0.80086, -0.15199]
512	4	LPIPS	+0.04232 [+0.02942, +0.05549]
512	4	DreamSim	+0.04502 [+0.02823, +0.06210]
512	4	CLIP-I	-0.02609 [-0.03777, -0.01427]
512	10	PSNR	-0.65111 [-0.98421, -0.33788]
512	10	LPIPS	+0.02955 [+0.02007, +0.03944]
512	10	DreamSim	+0.03454 [+0.02316, +0.04608]
512	10	CLIP-I	-0.02315 [-0.03368, -0.01277]
1024	10	PSNR	-0.15586 [-0.36917, +0.05727]
1024	10	LPIPS	+0.00378 [+0.00067, +0.00707]
1024	10	DreamSim	+0.00358 [+0.00101, +0.00601]
1024	10	CLIP-I	-0.00131 [-0.00523, +0.00253]

S4 Equal information budgets

The information budget includes 12 profile bits, header CRC16, length13 and body CRC16: 57 shared bits, plus 10 class bits for Ada. The arithmetic-code limits are $B - 57$ for WHOLE/NeST and $B - 67$ for Ada. The fixed budgets are $B = 620, 1212, 1302, 1940, 2248, 2577$ bits. WHOLE/NeST share maximum $(10, 0)$; Ada permits 1–256 tokens. Actual information fields and arithmetic streams are independently parsed at RX without PHY noise. The receiver cannot inspect unsent source tokens. This comparison removes wireless and MCS effects, but retains differences among tokenizer, prior, conditioning and decoder.

WHOLE and NeST have one deterministic output per source/budget. Ada uses the three fixed generation anchors 2026103001–2026103003, with the original per-source seed mechanism; metrics, not RGBs, are averaged. There are at most 15 000 logical outputs and 9000 source-level units. These are generation repeats, not new channel realizations.

All 15 000 logical outputs were accepted by the actual source receiver; no NULL or gray output occurs in this error-free diagnostic. New reconstruction requires 195 NeST/WHOLE receiver calls and 8733 Ada calls; 4466 NeST/WHOLE and 225 Ada input identities are reused. There are 8882 newly scored native pairs; 6118 logical positions reuse exact completed metric pairs. These counts are distinct from the number of information-transfer outcomes.

At every budget all five direct NeST–WHOLE intervals favor NeST. NeST–Ada PSNR remains positive throughout. DISTS favors NeST at 620/1212/1302/1940 bits and Ada at 2248/2577 bits. DreamSim favors NeST at 620/1212/1302 bits, crosses zero at 1940 bits, and favors Ada at 2248/2577 bits. LPIPS favors NeST through 2248 bits and crosses zero at 2577 bits. CLIP-I favors Ada at 620/1940/2577 bits, with intervals crossing zero at the other budgets. These mixed results remain part of the complete-system comparison. Increasing the budget from 1212 to 1302 bits raises mean source bits from 915.858 to 918.954 for WHOLE and from 1150.046 to 1239.950 for NeST. Thus NeST uses 89.904 of the extra 90 bits on average, while WHOLE uses 3.096; this is a source mean, not a guarantee for every image.

S4.1 Complete means and direct paired differences

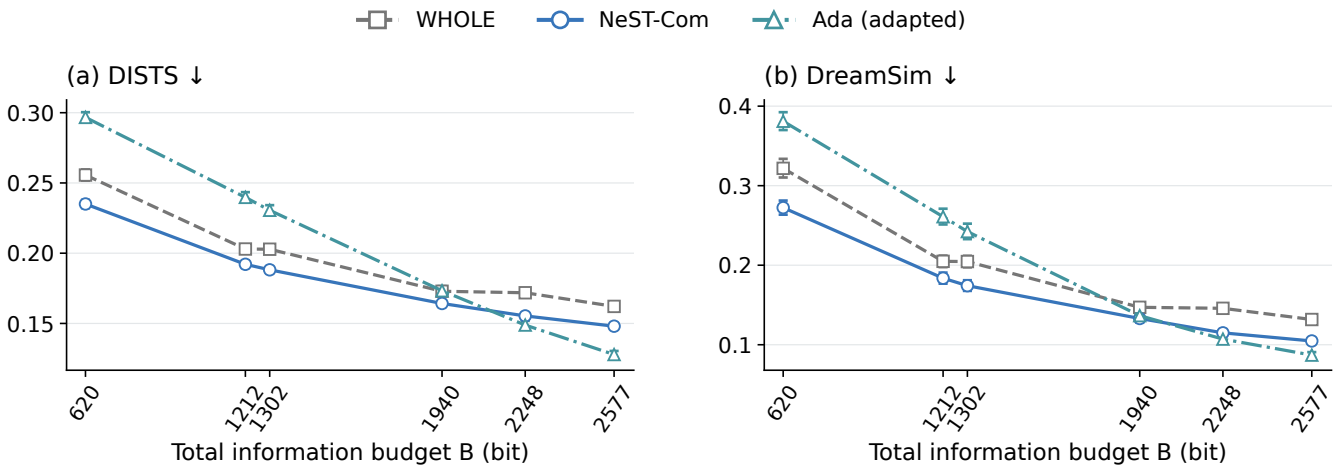


Figure S2: Error-free information-budget comparison on common500. Total bits include all registered information overhead. Numeric budget positions retain their unequal spacing. Points and bars are equal-source means and pointwise 95% source-bootstrap intervals. Ada's three generation repeats are averaged within source; they are not channel repeats. Lower distances are favorable.

Bits	Method	Metric	Mean [95% CI]
620	Ada	PSNR	11.34660 [11.16969, 11.53009]
620	Ada	LPIPS	0.54964 [0.54237, 0.55683]
620	Ada	DreamSim	0.38102 [0.36997, 0.39238]
620	Ada	CLIP-I	0.78893 [0.78097, 0.79684]
620	Ada	DISTS	0.29666 [0.29296, 0.30036]
1212	Ada	PSNR	13.06931 [12.86404, 13.27393]
1212	Ada	LPIPS	0.41291 [0.40557, 0.42011]
1212	Ada	DreamSim	0.26099 [0.25122, 0.27104]
1212	Ada	CLIP-I	0.82137 [0.81375, 0.82882]
1212	Ada	DISTS	0.23974 [0.23613, 0.24344]
1302	Ada	PSNR	13.38734 [13.18086, 13.59338]
1302	Ada	LPIPS	0.39382 [0.38649, 0.40102]
1302	Ada	DreamSim	0.24229 [0.23279, 0.25224]
1302	Ada	CLIP-I	0.83018 [0.82262, 0.83736]
1302	Ada	DISTS	0.23054 [0.22700, 0.23419]
1940	Ada	PSNR	16.06878 [15.80180, 16.34242]
1940	Ada	LPIPS	0.26649 [0.25947, 0.27363]
1940	Ada	DreamSim	0.13658 [0.13048, 0.14283]
1940	Ada	CLIP-I	0.86146 [0.85470, 0.86805]
1940	Ada	DISTS	0.17328 [0.17004, 0.17657]
2248	Ada	PSNR	17.74405 [17.45263, 18.03827]
2248	Ada	LPIPS	0.21299 [0.20638, 0.21971]
2248	Ada	DreamSim	0.10710 [0.10246, 0.11197]
2248	Ada	CLIP-I	0.87329 [0.86679, 0.87978]
2248	Ada	DISTS	0.14885 [0.14596, 0.15185]
2577	Ada	PSNR	19.52922 [19.23907, 19.81767]
2577	Ada	LPIPS	0.16729 [0.16181, 0.17293]
2577	Ada	DreamSim	0.08696 [0.08340, 0.09072]
2577	Ada	CLIP-I	0.88382 [0.87767, 0.88999]
2577	Ada	DISTS	0.12787 [0.12544, 0.13041]
620	NeST	PSNR	15.61746 [15.38069, 15.86297]
620	NeST	LPIPS	0.35487 [0.34755, 0.36210]
620	NeST	DreamSim	0.27236 [0.26357, 0.28127]

Bits	Method	Metric	Mean [95% CI]
620	NeST	CLIP-I	0.76969 [0.76034, 0.77914]
620	NeST	DISTS	0.23503 [0.23212, 0.23792]
1212	NeST	PSNR	17.41475 [17.17106, 17.67133]
1212	NeST	LPIPS	0.26496 [0.25874, 0.27117]
1212	NeST	DreamSim	0.18366 [0.17682, 0.19069]
1212	NeST	CLIP-I	0.82110 [0.81230, 0.82951]
1212	NeST	DISTS	0.19210 [0.18954, 0.19466]
1302	NeST	PSNR	17.65585 [17.40777, 17.91721]
1302	NeST	LPIPS	0.25492 [0.24880, 0.26113]
1302	NeST	DreamSim	0.17425 [0.16774, 0.18109]
1302	NeST	CLIP-I	0.82724 [0.81846, 0.83553]
1302	NeST	DISTS	0.18816 [0.18561, 0.19071]
1940	NeST	PSNR	19.04651 [18.79134, 19.31528]
1940	NeST	LPIPS	0.20235 [0.19702, 0.20764]
1940	NeST	DreamSim	0.13301 [0.12797, 0.13834]
1940	NeST	CLIP-I	0.85660 [0.84903, 0.86410]
1940	NeST	DISTS	0.16421 [0.16170, 0.16672]
2248	NeST	PSNR	19.63408 [19.36882, 19.90865]
2248	NeST	LPIPS	0.18322 [0.17818, 0.18827]
2248	NeST	DreamSim	0.11495 [0.11055, 0.11952]
2248	NeST	CLIP-I	0.86890 [0.86184, 0.87572]
2248	NeST	DISTS	0.15536 [0.15290, 0.15784]
2577	NeST	PSNR	20.19536 [19.92833, 20.47116]
2577	NeST	LPIPS	0.16577 [0.16123, 0.17038]
2577	NeST	DreamSim	0.10475 [0.10098, 0.10872]
2577	NeST	CLIP-I	0.87698 [0.87027, 0.88356]
2577	NeST	DISTS	0.14804 [0.14570, 0.15046]
620	WHOLE	PSNR	15.12224 [14.87019, 15.38250]
620	WHOLE	LPIPS	0.39246 [0.38283, 0.40199]
620	WHOLE	DreamSim	0.32180 [0.31030, 0.33372]
620	WHOLE	CLIP-I	0.74806 [0.73814, 0.75801]
620	WHOLE	DISTS	0.25563 [0.25172, 0.25960]
1212	WHOLE	PSNR	16.82131 [16.59007, 17.06388]
1212	WHOLE	LPIPS	0.28808 [0.28216, 0.29399]
1212	WHOLE	DreamSim	0.20486 [0.19764, 0.21207]
1212	WHOLE	CLIP-I	0.80744 [0.79868, 0.81586]
1212	WHOLE	DISTS	0.20297 [0.20052, 0.20552]
1302	WHOLE	PSNR	16.83227 [16.60005, 17.07610]
1302	WHOLE	LPIPS	0.28783 [0.28187, 0.29378]
1302	WHOLE	DreamSim	0.20460 [0.19737, 0.21184]
1302	WHOLE	CLIP-I	0.80770 [0.79894, 0.81618]
1302	WHOLE	DISTS	0.20283 [0.20033, 0.20541]
1940	WHOLE	PSNR	18.55591 [18.31286, 18.81036]
1940	WHOLE	LPIPS	0.21914 [0.21398, 0.22429]
1940	WHOLE	DreamSim	0.14707 [0.14173, 0.15258]
1940	WHOLE	CLIP-I	0.84737 [0.83942, 0.85502]
1940	WHOLE	DISTS	0.17290 [0.17051, 0.17533]
2248	WHOLE	PSNR	18.63777 [18.38451, 18.90215]
2248	WHOLE	LPIPS	0.21754 [0.21220, 0.22296]
2248	WHOLE	DreamSim	0.14580 [0.14038, 0.15132]
2248	WHOLE	CLIP-I	0.84835 [0.84040, 0.85602]
2248	WHOLE	DISTS	0.17184 [0.16935, 0.17438]
2577	WHOLE	PSNR	19.31282 [19.02087, 19.60883]
2577	WHOLE	LPIPS	0.19872 [0.19218, 0.20525]
2577	WHOLE	DreamSim	0.13172 [0.12594, 0.13763]
2577	WHOLE	CLIP-I	0.85774 [0.84973, 0.86557]

Bits	Method	Metric	Mean [95% CI]
2577	WHOLE	DISTS	0.16213 [0.15914, 0.16511]

Bits	Comparison	Metric	Difference [95% CI]
620	NeST–WHOLE	PSNR	+0.49521 [0.43094, 0.56242]
620	NeST–WHOLE	LPIPS	-0.03759 [-0.04141, -0.03380]
620	NeST–WHOLE	DreamSim	-0.04944 [-0.05531, -0.04362]
620	NeST–WHOLE	CLIP-I	+0.02163 [0.01652, 0.02682]
620	NeST–WHOLE	DISTS	-0.02059 [-0.02269, -0.01849]
620	NeST–Ada	PSNR	+4.27086 [4.10364, 4.44710]
620	NeST–Ada	LPIPS	-0.19476 [-0.20160, -0.18812]
620	NeST–Ada	DreamSim	-0.10866 [-0.11918, -0.09816]
620	NeST–Ada	CLIP-I	-0.01924 [-0.02717, -0.01122]
620	NeST–Ada	DISTS	-0.06163 [-0.06509, -0.05818]
620	WHOLE–Ada	PSNR	+3.77565 [3.59676, 3.96826]
620	WHOLE–Ada	LPIPS	-0.15718 [-0.16557, -0.14903]
620	WHOLE–Ada	DreamSim	-0.05922 [-0.07128, -0.04705]
620	WHOLE–Ada	CLIP-I	-0.04087 [-0.04921, -0.03263]
620	WHOLE–Ada	DISTS	-0.04103 [-0.04514, -0.03690]
1212	NeST–WHOLE	PSNR	+0.59345 [0.54574, 0.64438]
1212	NeST–WHOLE	LPIPS	-0.02311 [-0.02455, -0.02166]
1212	NeST–WHOLE	DreamSim	-0.02119 [-0.02371, -0.01875]
1212	NeST–WHOLE	CLIP-I	+0.01366 [0.01008, 0.01720]
1212	NeST–WHOLE	DISTS	-0.01088 [-0.01186, -0.00992]
1212	NeST–Ada	PSNR	+4.34544 [4.17204, 4.52565]
1212	NeST–Ada	LPIPS	-0.14794 [-0.15327, -0.14272]
1212	NeST–Ada	DreamSim	-0.07733 [-0.08526, -0.06928]
1212	NeST–Ada	CLIP-I	-0.00027 [-0.00670, 0.00590]
1212	NeST–Ada	DISTS	-0.04764 [-0.05066, -0.04462]
1212	WHOLE–Ada	PSNR	+3.75200 [3.58339, 3.92664]
1212	WHOLE–Ada	LPIPS	-0.12483 [-0.13038, -0.11947]
1212	WHOLE–Ada	DreamSim	-0.05614 [-0.06466, -0.04776]
1212	WHOLE–Ada	CLIP-I	-0.01392 [-0.02067, -0.00734]
1212	WHOLE–Ada	DISTS	-0.03677 [-0.03992, -0.03365]
1302	NeST–WHOLE	PSNR	+0.82358 [0.76712, 0.88222]
1302	NeST–WHOLE	LPIPS	-0.03292 [-0.03454, -0.03131]
1302	NeST–WHOLE	DreamSim	-0.03035 [-0.03337, -0.02733]
1302	NeST–WHOLE	CLIP-I	+0.01953 [0.01574, 0.02327]
1302	NeST–WHOLE	DISTS	-0.01466 [-0.01575, -0.01359]
1302	NeST–Ada	PSNR	+4.26851 [4.10141, 4.44272]
1302	NeST–Ada	LPIPS	-0.13891 [-0.14393, -0.13391]
1302	NeST–Ada	DreamSim	-0.06804 [-0.07541, -0.06074]
1302	NeST–Ada	CLIP-I	-0.00294 [-0.00857, 0.00249]
1302	NeST–Ada	DISTS	-0.04238 [-0.04529, -0.03948]
1302	WHOLE–Ada	PSNR	+3.44493 [3.28074, 3.61234]
1302	WHOLE–Ada	LPIPS	-0.10599 [-0.11125, -0.10079]
1302	WHOLE–Ada	DreamSim	-0.03769 [-0.04597, -0.02959]
1302	WHOLE–Ada	CLIP-I	-0.02247 [-0.02877, -0.01625]
1302	WHOLE–Ada	DISTS	-0.02772 [-0.03074, -0.02472]
1940	NeST–WHOLE	PSNR	+0.49060 [0.45242, 0.52988]
1940	NeST–WHOLE	LPIPS	-0.01680 [-0.01777, -0.01581]
1940	NeST–WHOLE	DreamSim	-0.01406 [-0.01570, -0.01247]
1940	NeST–WHOLE	CLIP-I	+0.00923 [0.00637, 0.01217]
1940	NeST–WHOLE	DISTS	-0.00870 [-0.00955, -0.00786]
1940	NeST–Ada	PSNR	+2.97773 [2.80999, 3.15216]
1940	NeST–Ada	LPIPS	-0.06414 [-0.06812, -0.06032]

Bits	Comparison	Metric	Difference [95% CI]
1940	NeST–Ada	DreamSim	-0.00357 [-0.00817, 0.00074]
1940	NeST–Ada	CLIP-I	-0.00487 [-0.00963, -0.00018]
1940	NeST–Ada	DISTS	-0.00907 [-0.01158, -0.00668]
1940	WHOLE–Ada	PSNR	+2.48713 [2.31261, 2.66864]
1940	WHOLE–Ada	LPIPS	-0.04735 [-0.05157, -0.04334]
1940	WHOLE–Ada	DreamSim	+0.01049 [0.00563, 0.01511]
1940	WHOLE–Ada	CLIP-I	-0.01409 [-0.01920, -0.00923]
1940	WHOLE–Ada	DISTS	-0.00037 [-0.00304, 0.00222]
2248	NeST–WHOLE	PSNR	+0.99631 [0.94081, 1.05089]
2248	NeST–WHOLE	LPIPS	-0.03432 [-0.03578, -0.03286]
2248	NeST–WHOLE	DreamSim	-0.03085 [-0.03332, -0.02848]
2248	NeST–WHOLE	CLIP-I	+0.02054 [0.01698, 0.02425]
2248	NeST–WHOLE	DISTS	-0.01648 [-0.01751, -0.01546]
2248	NeST–Ada	PSNR	+1.89003 [1.74362, 2.04058]
2248	NeST–Ada	LPIPS	-0.02977 [-0.03302, -0.02669]
2248	NeST–Ada	DreamSim	+0.00784 [0.00454, 0.01100]
2248	NeST–Ada	CLIP-I	-0.00439 [-0.00907, 0.00014]
2248	NeST–Ada	DISTS	+0.00651 [0.00445, 0.00847]
2248	WHOLE–Ada	PSNR	+0.89372 [0.73796, 1.05249]
2248	WHOLE–Ada	LPIPS	+0.00455 [0.00100, 0.00805]
2248	WHOLE–Ada	DreamSim	+0.03869 [0.03458, 0.04270]
2248	WHOLE–Ada	CLIP-I	-0.02493 [-0.02997, -0.02010]
2248	WHOLE–Ada	DISTS	+0.02299 [0.02077, 0.02515]
2577	NeST–WHOLE	PSNR	+0.88254 [0.80859, 0.95733]
2577	NeST–WHOLE	LPIPS	-0.03295 [-0.03563, -0.03024]
2577	NeST–WHOLE	DreamSim	-0.02698 [-0.02991, -0.02409]
2577	NeST–WHOLE	CLIP-I	+0.01924 [0.01555, 0.02311]
2577	NeST–WHOLE	DISTS	-0.01409 [-0.01534, -0.01283]
2577	NeST–Ada	PSNR	+0.66614 [0.55643, 0.78015]
2577	NeST–Ada	LPIPS	-0.00152 [-0.00390, 0.00071]
2577	NeST–Ada	DreamSim	+0.01779 [0.01549, 0.02007]
2577	NeST–Ada	CLIP-I	-0.00685 [-0.01099, -0.00266]
2577	NeST–Ada	DISTS	+0.02017 [0.01851, 0.02177]
2577	WHOLE–Ada	PSNR	-0.21640 [-0.34068, -0.09134]
2577	WHOLE–Ada	LPIPS	+0.03143 [0.02804, 0.03476]
2577	WHOLE–Ada	DreamSim	+0.04477 [0.04100, 0.04860]
2577	WHOLE–Ada	CLIP-I	-0.02608 [-0.03122, -0.02109]
2577	WHOLE–Ada	DISTS	+0.03426 [0.03241, 0.03607]

S4.2 Actual source lengths and endpoint saturation

Bits	Method	Source bits	Padding bits	Full / 500	NULL / 500
620	Ada	547.58	5.42	0	0
1212	Ada	1139.70	5.30	0	0
1302	Ada	1229.54	5.46	0	0
1940	Ada	1866.82	6.18	4	0
2248	Ada	2170.61	10.39	10	0
2577	Ada	2477.89	32.11	75	0
620	NeST	558.28	4.72	0	0
1212	NeST	1150.05	4.95	0	0
1302	NeST	1239.95	5.05	0	0
1940	NeST	1878.36	4.64	0	0
2248	NeST	2186.54	4.46	0	0
2577	NeST	2515.72	4.28	0	0

Bits	Method	Source bits	Padding bits	Full / 500	NULL / 500
620	WHOLE	427.80	135.20	0	0
1212	WHOLE	915.86	239.14	0	0
1302	WHOLE	918.95	326.05	0	0
1940	WHOLE	1566.43	316.57	0	0
2248	WHOLE	1599.71	591.29	0	0
2577	WHOLE	1958.14	561.86	0	0

Full means all 10 NeST/WHOLE scales or all 256 Ada tokens. These counts are computed once per source, not once per Ada repeat. Padding after a complete representation is not failure to fit a prefix. Source bits exclude the separately stated information overhead.

S5 Fixed-policy channel threshold

The 10-dB policies of WHOLE, NeST and Ada remain frozen at actual SNRs 8, 8.5, 9, 9.5 and 10 dB. Each cell has 100 sources and 20 new AWGN vectors, hence 2000 actual complete frames and 30 000 frames in total. The source's original repeat-zero payload, waveform, session, counter and scrambler identity are retained; only noise changes. RX uses the true noise variance. Ada's original repeat-zero generation identity remains fixed across noise and SNR.

All 30 000 real receptions and both distance metrics have completed. There are 17 497 accepted outcomes and 12 503 body/profile-parse rejections, with no header rejection, unexpected accepted source input, source-decoding rejection or NULL. No additional image generation is needed: accepted prefixes and frozen generation identities match existing outputs. Scoring required 88 new native gray-reference pairs for each distance; all other pairs were reused after identity verification.

All methods reject all 2000 receptions at 8 dB. At 8.5 dB their rejection rates are 93.05–94.00%, at 9 dB they are 13.50–15.75%, and at 9.5 dB they are 0.10–0.20%. No rejection is observed at 10 dB. Source-cluster intervals retain all 20 noise repetitions within each sampled source. This finite scan describes the tested fixed policies, not low-error-rate operation or general channel robustness.

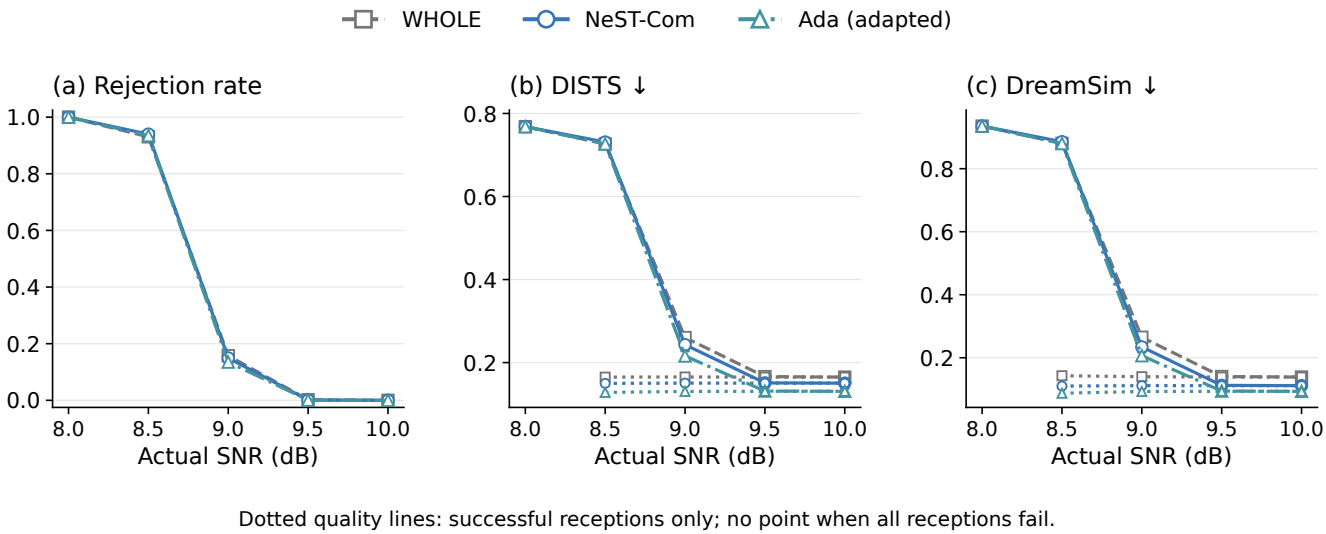


Figure S3: Frozen 10-dB policies evaluated at five actual SNRs. Points and bars retain all receptions and show pointwise 95% source-bootstrap intervals. Dotted quality curves describe successful receptions only and use a changing event denominator; no conditional point exists at 8 dB.

Method	SNR	Metric	Overall mean [95% CI]
Ada	8	Rejection rate	1.00000 [1.00000, 1.00000]
Ada	8	DISTS	0.76824 [0.76057, 0.77563]

Method	SNR	Metric	Overall mean [95% CI]
Ada	8	DreamSim	0.93515 [0.92703, 0.94316]
Ada	8.5	Rejection rate	0.93500 [0.92200, 0.94700]
Ada	8.5	DISTS	0.72680 [0.71557, 0.73771]
Ada	8.5	DreamSim	0.87996 [0.86721, 0.89279]
Ada	9	Rejection rate	0.13500 [0.11950, 0.15100]
Ada	9	DISTS	0.21603 [0.20456, 0.22794]
Ada	9	DreamSim	0.20653 [0.19141, 0.22233]
Ada	9.5	Rejection rate	0.00100 [0.00000, 0.00250]
Ada	9.5	DISTS	0.13146 [0.12526, 0.13813]
Ada	9.5	DreamSim	0.09424 [0.08616, 0.10328]
Ada	10	Rejection rate	0.00000 [0.00000, 0.00000]
Ada	10	DISTS	0.13079 [0.12463, 0.13742]
Ada	10	DreamSim	0.09341 [0.08534, 0.10239]
NeST	8	Rejection rate	1.00000 [1.00000, 1.00000]
NeST	8	DISTS	0.76824 [0.76057, 0.77563]
NeST	8	DreamSim	0.93515 [0.92703, 0.94316]
NeST	8.5	Rejection rate	0.94000 [0.92950, 0.95000]
NeST	8.5	DISTS	0.73115 [0.72148, 0.74057]
NeST	8.5	DreamSim	0.88560 [0.87443, 0.89682]
NeST	9	Rejection rate	0.14950 [0.13200, 0.16750]
NeST	9	DISTS	0.24294 [0.23094, 0.25502]
NeST	9	DreamSim	0.23428 [0.21806, 0.25085]
NeST	9.5	Rejection rate	0.00100 [0.00000, 0.00250]
NeST	9.5	DISTS	0.15110 [0.14561, 0.15663]
NeST	9.5	DreamSim	0.11231 [0.10429, 0.12089]
NeST	10	Rejection rate	0.00000 [0.00000, 0.00000]
NeST	10	DISTS	0.15047 [0.14502, 0.15601]
NeST	10	DreamSim	0.11149 [0.10345, 0.11997]
WHOLE	8	Rejection rate	1.00000 [1.00000, 1.00000]
WHOLE	8	DISTS	0.76824 [0.76057, 0.77563]
WHOLE	8	DreamSim	0.93515 [0.92703, 0.94316]
WHOLE	8.5	Rejection rate	0.93050 [0.91750, 0.94250]
WHOLE	8.5	DISTS	0.72658 [0.71546, 0.73707]
WHOLE	8.5	DreamSim	0.88020 [0.86695, 0.89269]
WHOLE	9	Rejection rate	0.15750 [0.14300, 0.17200]
WHOLE	9	DISTS	0.26020 [0.25049, 0.27014]
WHOLE	9	DreamSim	0.26497 [0.25173, 0.27887]
WHOLE	9.5	Rejection rate	0.00200 [0.00050, 0.00400]
WHOLE	9.5	DISTS	0.16623 [0.15958, 0.17286]
WHOLE	9.5	DreamSim	0.14037 [0.12846, 0.15272]
WHOLE	10	Rejection rate	0.00000 [0.00000, 0.00000]
WHOLE	10	DISTS	0.16500 [0.15844, 0.17169]
WHOLE	10	DreamSim	0.13874 [0.12689, 0.15117]

Method	SNR	Success / 2000	Cond. DISTS	Cond. DreamSim
Ada	8	0	–	–
Ada	8.5	130	0.12820	0.08738
Ada	9	1730	0.13051	0.09304
Ada	9.5	1998	0.13079	0.09340
Ada	10	2000	0.13079	0.09341
NeST	8	0	–	–
NeST	8.5	120	0.14992	0.11011
NeST	9	1701	0.15061	0.11145
NeST	9.5	1998	0.15047	0.11151
NeST	10	2000	0.15047	0.11149

Method	SNR	Success / 2000	Cond. DISTS	Cond. DreamSim
WHOLE	8	0	–	–
WHOLE	8.5	139	0.16504	0.14297
WHOLE	9	1685	0.16541	0.13993
WHOLE	9.5	1996	0.16503	0.13878
WHOLE	10	2000	0.16500	0.13874

S6 Qualitative examples

Fig.3 uses sources 135 and 201 at $N = 1024$, 10 dB (catalog C10-016/C10-017), chosen after inspection of the complete candidate gallery to illustrate perceptual advantages. These are selected examples, not random or representative samples. The full IDs are `n04461696/ILSVRC2012_val_00026517_n04461696` and `n06596364/ILSVRC2012_val_00001019_n06596364`. NeST has lower DISTS and DreamSim than WHOLE, SwinJSCC and Ada in both displayed realizations; this does not imply superiority in every metric or over the population. SwinJSCC/NeST PSNR is 21.96/19.99 dB and 19.75/16.82 dB, respectively. Ada has lower LPIPS on source 135 (0.1955 versus 0.2112) and higher CLIP-I on source 201 (0.6926 versus 0.6596). Full-precision scores and image identities are in `inputs/fig3_selected_20261102/`.

All images retain the full 256×256 field and each method's original first channel repetition: seed 6201 for WHOLE, NeST and Ada; seed 2001 for SwinJSCC, preserving historical method-specific noise. No failure or repetition was substituted. Metrics use the native float outputs; PNGs serve display only. The following panel shows the selected 7-dB television (index 84) and leather (index 143) advantage examples with complete method comparisons. For a complementary view, the next two sources were fixed before their images were opened: common500 N1024/10dB, ranked by each source's mean DISTS gain of Token over WHOLE, with stable source-index tie breaking. We use the package's median-rank source 209 and minimum-gain source 39, without a success filter. Display is the first original repetition (noise 6201 for both methods). Source209 is `n04509417/ILSVRC2012_val_00035733_n04509417`; source 39 is `n02097130/ILSVRC2012_val_00048087_n02097130`. This is a fixed median-rank and DISTS-degradation comparison. Source39 worsens in DISTS but improves in DreamSim, illustrating disagreement between distances.

S7 Unified processing costs

S7.1 Corrected resource protocol for the main comparisons

The resource audit uses the same first 32 common500 IDs at $N = 1024$, 10 dB, batch one, with two warmups and three measurements per source and direction. All five GPU implementations run serially on the same physical RTX PRO 6000. Every method uses CPU affinity [38,40], which selects distinct physical cores 19 and 20 of socket 0, with two intra-op and two inter-op threads. BPG uses these CPU cores without a GPU. The numerical routes, weights, policies, source order and observed RX symbols remain frozen. Every emitted waveform and reconstructed image matches its historical event hash exactly. Ada's measurements below document our adaptation only and are withheld from the main timing comparison for the implementation reasons given below.

TX starts with canonical RGB in memory and ends with actual symbols; RX starts with received symbols in memory and ends with RGB. GPU synchronization brackets each span. Tokenization, condition extraction, CDF construction, arithmetic coding, frame fitting, link processing and generation are included. Model loading, input loading, final-output hashing, saved logs and quality scoring are outside the timer. BPG retains source search, codec invocation and RAM serialization. No cached source tokens or reconstructions replace work inside a timed span. TX+RX is the sum of processing spans.

The previous affinity [38,39] selected two SMT siblings of one physical core, although the earlier manuscript called them two CPU cores; its intra-op pool had six threads. This was the same allocation for all methods. The corrected run replaces the available main timing cells; Ada's cells remain unreported rather than presenting its serial adaptation as an author-aligned implementation. The initial GPU0 session was interrupted by a foreign GPU diagnostic process; all GPU0 timings are excluded. GPU6 supplies every final GPU cell, and only the completed, exactly matching CPU-only BPG cells are reused. Prior resource use and interrupted-call ledgers are retained. No foreign process or GPU clock setting was changed.

The table gives equal-source means in milliseconds. Brackets for total time are pointwise 95% source-bootstrap intervals (10 000 resamples, fixed seed 2026110201). They describe source variation during this session, not all possible system

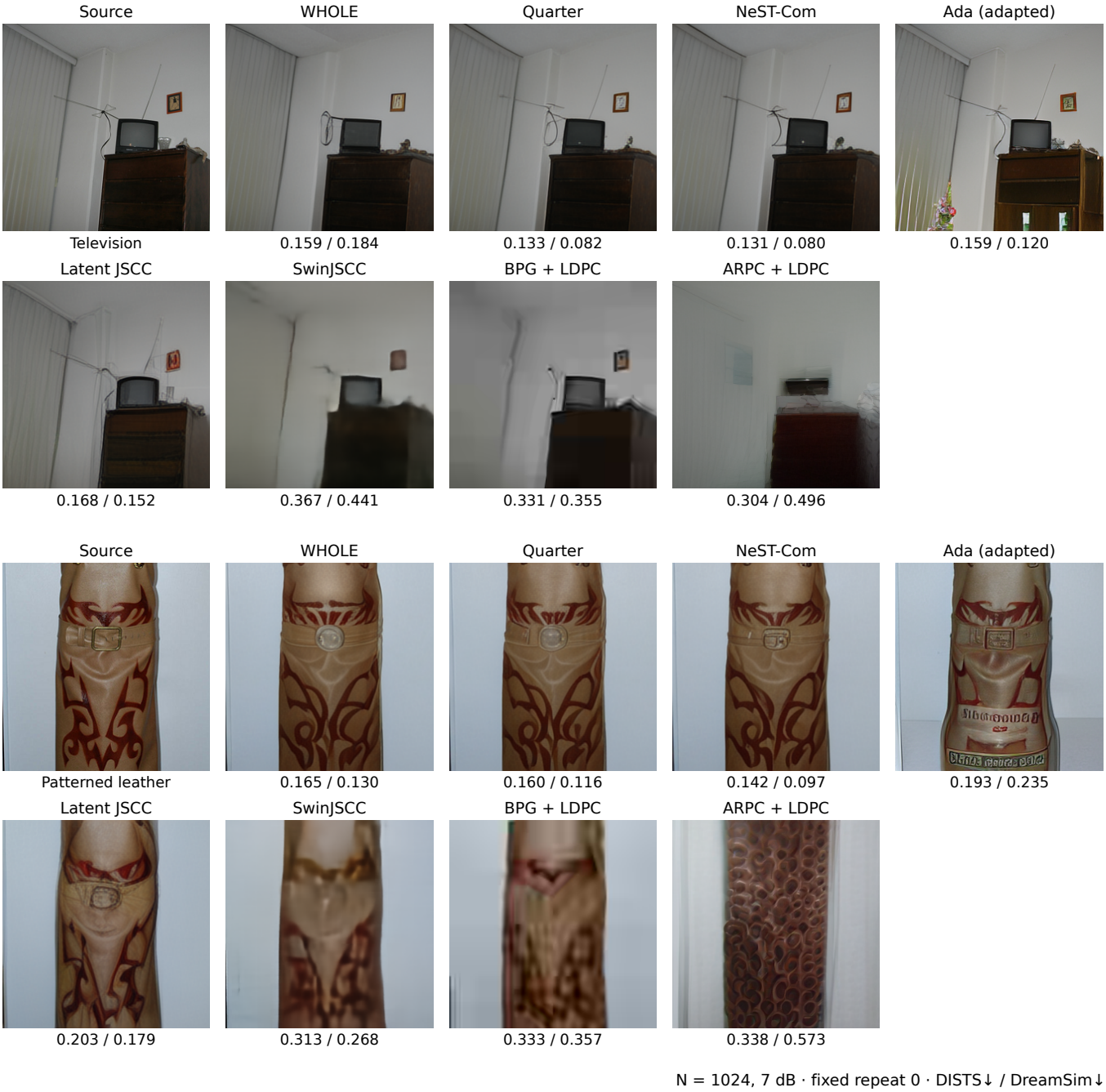


Figure S4: Complete method comparisons for the television and leather advantage examples. Full256×256 fields are retained.



Figure S5: Median-rank (top) and minimum DISTS-gain (bottom) sources at N1024/10dB. Columns are source, WHOLE and NeST. Values are DISTS / DreamSim for the displayed first repetition, lower favorable.

load. The six main methods each complete 96 measured RX calls successfully. Full call times, source means, median/p95 and resource telemetry are in `inputs/timing_fairness_20261102/` and the accompanying result directory.

System	TX (ms)	RX (ms)	Total (ms) [95% CI]
WHOLE	215.29	366.42	581.71 [576.77, 586.78]
NeST	248.44	403.99	652.43 [647.45, 657.51]
Ada (audit only)	7653.77	7888.77	15542.54 [15525.31, 15560.32]
Latent	15.28	23.93	39.21 [39.20, 39.22]
Swin	21.01	18.16	39.17 [39.07, 39.30]
BPG	2209.97	21.28	2231.26 [2163.61, 2298.68]

S7.2 Matched granularity and audit sensitivity

The matched grid fixes ceiling $(8, 126)$, model, MCS and fitting rule. Whole/Quarter/Eighth TX are measured in the same resource session; Token reuses the identical main NeST TX cell. The main WHOLE policy has a different ceiling and is not substituted for the matched Whole cell. Quality values and all received inputs are unchanged.

Grid	TX (ms)	95% source interval
Whole	185.20	[184.76, 185.67]
Quarter	215.84	[215.33, 216.41]
Eighth	217.76	[217.18, 218.48]
Token	248.44	[246.78, 251.06]

The frozen Ada adaptation evaluates probabilities serially for all 256 source positions at TX. It executes conditional and unconditional branches, although only the conditional branch supplies entropy CDFs, and retains its FP32 math-

attention route. RX decodes the prefix and continues generation along the same context. NeST predicts maps scale by scale. These model and implementation choices remain part of the measured costs.

Ada-TokenCom's published Section IV-E instead describes a single encoder prefill and `torch.compile`. Its Table IV reports 19 ms encoding and 1012 ms decoding on an RTX 5090 and Xeon Platinum 8260. The reported boundary comprises tokenizer, AR probability estimation, softmax and arithmetic coding, whereas our spans additionally include class prediction and digital-link processing. The paper also describes repeated full-prefix evaluation at RX to preserve entropy-model numerical consistency. Thus, neither its published numbers nor our serial-adaptation numbers establish a matched end-to-end speed comparison. In particular, the serial TX loop and unused TX branch are implementation overheads, not requirements of Ada's method. The published latency section does not specify enough numerical detail to assume that a different precision is responsible for the discrepancy.

An auxiliary diagnostic uses only the first two registered sources. It defers Ada's redundant per-token finite-logit audit until after timing and removes unused CDF audit hashes; for NeST it removes an unused completed-token hash. CRC, length/parser checks, arithmetic canonical re-encoding, actual generation and final output identity remain. The NeST TX path is unchanged and serves as a negative control. The following means compare this diagnostic with the original-audit measurements on the same two sources. Their session variation and tiny sample preclude replacing the 32-source results or subtracting these differences from them.

System	Span	Original audits	Diagnostic	Difference (ms)
NeST	TX	246.78	248.51	+1.72
NeST	RX	406.41	404.05	-2.37
Ada (audit only)	TX	7637.67	7664.81	+27.14
Ada (audit only)	RX	7872.74	7688.25	-184.49

The measurements characterize the frozen implementations on a common resource allocation. They do not measure equally optimized kernels or an architecture-independent speed limit. Ada's raw measurements remain audit evidence only, with no NeST/Ada speedup claim. All diagnostic outcomes, including the unchanged-path control, are retained.

S7.3 Prefill compatibility diagnostic

Two fixed sources from the preceding diagnostic are reused for four full-prefix probability calls: conditional-only and the original two-branch calculation for each source. We preserve FP32, math attention, weights and the exact integer-CDF conversion. Existing source tokens, predicted classes, 256 CDF hashes and 256 independently terminated streams per source provide the reference; no classifier, tokenizer, image decoder, PHY or quality evaluation is rerun. The table counts exact matches to the frozen incremental route.

Source index	Branches	CDF matches	Stream matches
0	1	0/256	4/256
0	2	0/256	4/256
1	1	0/256	2/256
1	2	0/256	2/256

The full-prefix route is not an exact replacement for the frozen incremental probability path on these inputs. It requires a separately validated transmitter/receiver numerical route rather than substituting new probabilities into old bitstreams. The following study validates such a paired route; the preceding diagnostic and original streams remain unchanged.

S7.4 Separately qualified compiled Ada implementation

We replace the serial encoder with a single conditional causal prefill and compile the forward operation with `torch.compile`. At RX, the conditional CDF is reconstructed from the received prefix alone using the same tensor shape and strict causal mask; unknown future inputs are public zeros. The generated tail follows the actual recovered context, using the original CFG=4, temperature, filtering and sampling rules. Entropy CDFs use only the conditional branch. We retain FP32, math attention, disabled TF32, the same weights, classifier, tokenizer, image decoder, fixed MCS and candidate set. The compiler may change floating-point execution, so this is a new numerical implementation, not a bit-exact substitution for the original quality-evaluation route.

Two fixed sources first pass exact recovery of all 512 tokens and 512 integer CDFs, plus two fixed-prefix tail checks. That successful qualification is reused, without repeating its model calls. The formal study then performs fresh RGB-to-symbol TX and independent symbol-to-RGB RX on the same 32 sources, with two warmups and three measurements per direction. RX has no access to source pixels or TX witness files. All 160 received prefixes, decoded classes and endpoints match the external TX witness after RX has exited; all 96 measured receptions succeed. Repeated outputs are deterministic within the new implementation.

The following timings include class prediction and actual digital-link processing at the same synchronized boundaries as the preceding resource audit. Model loading and cold compilation are excluded from latency, but remain charged to the resource ledger. Compilation uses eight CPU cores; each timed span returns to the same two distinct cores and 2/2 thread pools. No main image-quality experiment or metric evaluation is rerun. The 10 000-by-32 bootstrap index bank and source order are reused exactly.

Implementation span	Mean (ms)	95% source interval
TX	114.55	[114.15, 114.93]
RX	2949.76	[2945.89, 2953.91]
TOTAL	3064.30	[3060.46, 3068.37]

The new route matches 0/160 historical emitted waveforms and 10/160 historical reconstructed images exactly. Its latency is therefore reported separately from the original-quality Ada row; it does not replace those quality arrays or establish a NeST/Ada algorithmic speedup. The serial audit cost is not an intrinsic Ada requirement. The authors' published timing also uses different hardware and a different processing boundary, so its 19/1012 ms values are not substituted into the local comparison.

Three unsuccessful compilation attempts (missing development headers, parent and compiler-child module-name collisions) precede the formal run. They produced no timed TX/RX results and are retained with their actual nonzero exits and resource use. Private matching Python headers and the existing child-namespace binding repair the environment without changing the qualified model calculation. These failures are not included in per-image processing times. All study attempts remain charged against the original absolute deadline and the separately authorized resource extension.

S7.5 Historical E5 resource protocol

The historical 7-dB, N512, ARPC and grid-RX groups below were not rerun. For completeness, the original N1024/10 groups are also retained and labeled historical. They use affinity [38,39] (two SMT siblings of one physical core), six intra-op and two inter-op threads. These values are not combined with the corrected-resource main tables.

Timing uses the first 32 common500 IDs, batch one, two warmups and three measurements per source, one actual RTX PRO 6000 and two SMT siblings of one physical CPU core. Original numerical routes are retained; six library compute threads are confined to those two logical CPUs. Model loading, disk I/O, logging and quality scoring are excluded. TX starts with in-memory canonical RGB and ends with actual transmitted symbols. RX starts with in-memory received symbols and ends with canonical RGB. GPU boundary synchronization surrounds each measured span; TX+RX is a processing sum, not radio propagation latency.

The measured operations include tokenizer, condition extraction, CDF/arithmetic coding, finalization/fitting, paid packaging, LDPC/QAM and real receiver PHY, independent source decoding, completion and image decoding. Ada includes ConvNeXt prediction; ARPC includes caption generation; BPG includes its actual search and codec invocation. BPG is CPU-only on the same host and CPU quota; necessary codec serialization uses RAM files. Fresh timing does not substitute cached source tokens or reconstructions. Successful-RX conditional time is reported separately from fast rejection. Means first average within source; median/p95 use measured calls with linear quantiles.

All 16 groups contribute 3072 measured boundary spans after 2048 warmups. ARPC includes its original model runtime and CPU-PHY interprocess transfer within the same SMT-pair affinity. BPG includes codec invocation and source search.

S7.6 Historical measured groups

Values below are milliseconds. Means average the three measurements within each of 32 sources; median and p95 are calculated from the 96 measured calls. Total pairs the TX and RX spans for the same source and repetition. The success

count refers to measured RX calls; conditional RX means exclude rejected receptions and are not substituted for overall RX means.

System	N	SNR	Span	Mean (ms)	Median	p95
WHOLE	1024	7	TX	263.35	262.01	270.94
WHOLE	1024	7	RX	375.26	374.77	379.50
WHOLE	1024	7	total	638.61	637.56	646.96
NeST	1024	7	TX	362.09	360.53	370.70
NeST	1024	7	RX	390.86	390.64	394.25
NeST	1024	7	total	752.95	751.30	761.83
Ada	1024	7	TX	7546.57	7537.00	7627.76
Ada	1024	7	RX	7873.85	7864.66	7932.79
Ada	1024	7	total	15420.41	15404.73	15537.01
WHOLE	1024	10	TX	213.83	213.37	216.53
WHOLE	1024	10	RX	369.90	371.15	387.29
WHOLE	1024	10	total	583.73	585.23	602.20
NeST	1024	10	TX	248.38	247.09	251.45
NeST	1024	10	RX	411.64	401.40	431.95
NeST	1024	10	total	660.02	649.47	678.57
Ada	1024	10	TX	7795.61	7790.73	7854.45
Ada	1024	10	RX	7626.11	7623.30	7667.14
Ada	1024	10	total	15421.72	15412.42	15498.35
WHOLE	512	10	TX	157.33	157.00	159.30
WHOLE	512	10	RX	325.48	325.10	328.35
WHOLE	512	10	total	482.81	482.32	485.94
NeST	512	10	TX	163.34	163.14	164.79
NeST	512	10	RX	348.89	348.50	352.27
NeST	512	10	total	512.22	511.81	515.45
Ada	512	10	TX	7378.20	7373.40	7423.35
Ada	512	10	RX	7789.73	7785.50	7836.21
Ada	512	10	total	15167.93	15159.94	15239.95
ARPC	1024	10	TX	661.72	653.02	732.59
ARPC	1024	10	RX	797.03	796.61	803.32
ARPC	1024	10	total	1458.76	1448.62	1531.35
Swin	1024	10	TX	20.53	20.47	20.67
Swin	1024	10	RX	17.61	17.60	17.73
Swin	1024	10	total	38.14	38.08	38.41
Latent	1024	10	TX	15.42	15.41	15.47
Latent	1024	10	RX	24.10	24.05	24.21
Latent	1024	10	total	39.51	39.47	39.64
BPG	1024	10	TX	2441.98	2457.79	2827.31
BPG	1024	10	RX	31.71	31.58	33.13
BPG	1024	10	total	2473.69	2489.51	2859.12
Grid Whole	1024	10	TX	183.50	182.95	186.63
Grid Whole	1024	10	RX	378.16	380.12	393.96
Grid Whole	1024	10	total	561.66	569.55	578.15
Grid Quarter	1024	10	TX	214.09	213.20	216.88
Grid Quarter	1024	10	RX	385.21	383.68	388.92
Grid Quarter	1024	10	total	599.30	597.40	609.10
Grid Eighth	1024	10	TX	217.25	216.96	219.00
Grid Eighth	1024	10	RX	388.82	386.04	414.90
Grid Eighth	1024	10	total	606.07	603.35	630.91

System	N	SNR	Success / 96	Conditional RX (ms)
WHOLE	1024	7	96	375.26
NeST	1024	7	96	390.86

System	N	SNR	Success / 96	Conditional RX (ms)
Ada	1024	7	96	7873.85
WHOLE	1024	10	96	369.90
NeST	1024	10	96	411.64
Ada	1024	10	96	7626.11
WHOLE	512	10	96	325.48
NeST	512	10	96	348.89
Ada	512	10	96	7789.73
ARPC	1024	10	96	797.03
Swin	1024	10	96	17.61
Latent	1024	10	96	24.10
BPG	1024	10	96	31.71
Grid Whole	1024	10	96	378.16
Grid Quarter	1024	10	96	385.21
Grid Eighth	1024	10	96	388.82

S7.7 Shared computation accounting

The matched grid comparison uses ceiling $(8, 126)$; its token arm reuses the identical NeST timing group at $N = 1024$, 10 dB. The main WHOLE timing group has its own family target and is not the matched-grid WHOLE control. The timed implementation constructs all candidate streams and scans the complete list for feasibility before selecting the longest one. Its recorded legacy field `length_checks` is the selected endpoint's descending rank, not the actual number of tests. The corrected accounting records 5/20/39/352 candidate tests for matched Whole/Quarter/Eighth/Token, separately from the early-stopping comparisons used by the cached quality control in S3. Both rules select the same endpoint; the timing includes this implementation's full scan.

Fresh source-probability traversals, CDF construction steps, independent finalizations, actual tests and endpoint ranks are in `source_fitting_accounting.csv`, derived from the bound code and per-trial witnesses. Stage-level instrumented model/PHY calls are in `model_call_accounting.csv`; they are not exhaustive operator counts. Counters include warmups and are not interchangeable with latency. No prior source-output cache replaces timed model work.

This software implementation includes actual CDF construction, independent arithmetic termination and real digital decoding. The result is a deployment-cost comparison on this host and these original runtimes, not an optimized-hardware throughput bound. The table reports the observed success counts and descriptive timing statistics.

S8 Transfer to Kodak24

Kodak uses the original RGB center 256×256 crop without resizing before cropping. ImageNet-calibrated policies, model components and receiver configuration remain fixed. WHOLE and Quarter are the prior complete-scale and quarter-grid systems, respectively; Token uses integer endpoints. Ada remains the paid predicted-class fixed-frame adaptation. SNR 4/10/19 each use three original events per image (seeds2001/2002/2003), giving 216 logical results per method. Native outputs and existing four metrics are reused under exact identities; DISTS is evaluated on all 269 unique source/output pairs. Token needs 213 new physical frames plus 3 exact reused frames, and 71 new unique formal reconstructions plus 1 reused reconstruction. All 864 four-method events are accepted; the independent source population is 24, not 72.

Token improves PSNR, DISTS and DreamSim over WHOLE at all three SNRs. Compared with Quarter, Token's DISTS/DreamSim intervals exclude zero at 10/19 dB and cross zero at 4 dB. Against Ada, Token improves DISTS/DreamSim at 4 dB; Ada improves DISTS at 10/19 dB, while DreamSim differences cross zero. The complete five-metric results follow; all intervals use the existing 10 000×24 source bank after source-level event averaging.

SNR	Method	Metric	Mean [pointwise 95% CI]
4	WHOLE	PSNR	18.22252 [17.19135, 19.28486]
4	WHOLE	DISTS	0.20677 [0.19716, 0.21593]
4	WHOLE	LPIPS	0.29033 [0.26444, 0.31649]
4	WHOLE	DreamSim	0.20424 [0.17572, 0.23450]
4	WHOLE	CLIP-I	0.81941 [0.79144, 0.84729]

SNR	Method	Metric	Mean [pointwise 95% CI]
10	WHOLE	PSNR	20.33540 [19.00247, 21.72279]
10	WHOLE	DISTS	0.16732 [0.15447, 0.17974]
10	WHOLE	LPIPS	0.20294 [0.17231, 0.23357]
10	WHOLE	DreamSim	0.12658 [0.10627, 0.14680]
10	WHOLE	CLIP-I	0.85987 [0.83716, 0.88284]
19	WHOLE	PSNR	22.64749 [21.42126, 23.92284]
19	WHOLE	DISTS	0.12954 [0.12145, 0.13784]
19	WHOLE	LPIPS	0.12688 [0.11148, 0.14242]
19	WHOLE	DreamSim	0.08077 [0.06859, 0.09424]
19	WHOLE	CLIP-I	0.89736 [0.87628, 0.91739]
4	Quarter	PSNR	18.61812 [17.58014, 19.68850]
4	Quarter	DISTS	0.19469 [0.18526, 0.20428]
4	Quarter	LPIPS	0.27138 [0.24587, 0.29674]
4	Quarter	DreamSim	0.17664 [0.15190, 0.20216]
4	Quarter	CLIP-I	0.83837 [0.81319, 0.86334]
10	Quarter	PSNR	21.00895 [19.79903, 22.26482]
10	Quarter	DISTS	0.15675 [0.14639, 0.16737]
10	Quarter	LPIPS	0.17623 [0.15438, 0.19836]
10	Quarter	DreamSim	0.11016 [0.09240, 0.12951]
10	Quarter	CLIP-I	0.86917 [0.84631, 0.89104]
19	Quarter	PSNR	22.87538 [21.59650, 24.19178]
19	Quarter	DISTS	0.12711 [0.11842, 0.13604]
19	Quarter	LPIPS	0.12304 [0.10683, 0.13951]
19	Quarter	DreamSim	0.07949 [0.06724, 0.09289]
19	Quarter	CLIP-I	0.89545 [0.87353, 0.91613]
4	Token	PSNR	18.78991 [17.77901, 19.83027]
4	Token	DISTS	0.19566 [0.18560, 0.20525]
4	Token	LPIPS	0.26054 [0.23491, 0.28639]
4	Token	DreamSim	0.17165 [0.14835, 0.19668]
4	Token	CLIP-I	0.84106 [0.81457, 0.86665]
10	Token	PSNR	21.12961 [19.93020, 22.36030]
10	Token	DISTS	0.15356 [0.14400, 0.16359]
10	Token	LPIPS	0.17156 [0.14997, 0.19328]
10	Token	DreamSim	0.10473 [0.08826, 0.12314]
10	Token	CLIP-I	0.88441 [0.86585, 0.90309]
19	Token	PSNR	23.05219 [21.77905, 24.34964]
19	Token	DISTS	0.12508 [0.11634, 0.13390]
19	Token	LPIPS	0.11866 [0.10295, 0.13449]
19	Token	DreamSim	0.07746 [0.06447, 0.09191]
19	Token	CLIP-I	0.89827 [0.87404, 0.92010]
4	Ada	PSNR	13.53319 [12.61366, 14.57350]
4	Ada	DISTS	0.25729 [0.24236, 0.27212]
4	Ada	LPIPS	0.45443 [0.42323, 0.48381]
4	Ada	DreamSim	0.30625 [0.26196, 0.35335]
4	Ada	CLIP-I	0.78734 [0.75053, 0.82311]
10	Ada	PSNR	19.92384 [18.48343, 21.42034]
10	Ada	DISTS	0.13142 [0.11836, 0.14511]
10	Ada	LPIPS	0.17394 [0.14562, 0.20325]
10	Ada	DreamSim	0.10152 [0.08147, 0.12556]
10	Ada	CLIP-I	0.88105 [0.85052, 0.90936]
19	Ada	PSNR	21.46358 [20.29288, 22.67017]
19	Ada	DISTS	0.11340 [0.10648, 0.12014]
19	Ada	LPIPS	0.14084 [0.12365, 0.15827]
19	Ada	DreamSim	0.08271 [0.07083, 0.09517]
19	Ada	CLIP-I	0.89815 [0.87737, 0.91827]

SNR	Comparison	Metric	Difference [paired 95% CI]
4	Token minus WHOLE	PSNR	+0.56739 [+0.39543, +0.74525]
4	Token minus WHOLE	DISTS	-0.01111 [-0.01626, -0.00593]
4	Token minus WHOLE	LPIPS	-0.02979 [-0.03607, -0.02319]
4	Token minus WHOLE	DreamSim	-0.03259 [-0.04669, -0.01907]
4	Token minus WHOLE	CLIP-I	+0.02165 [+0.01043, +0.03289]
10	Token minus WHOLE	PSNR	+0.79421 [+0.50913, +1.09128]
10	Token minus WHOLE	DISTS	-0.01376 [-0.01986, -0.00788]
10	Token minus WHOLE	LPIPS	-0.03138 [-0.04223, -0.02047]
10	Token minus WHOLE	DreamSim	-0.02185 [-0.03313, -0.01098]
10	Token minus WHOLE	CLIP-I	+0.02454 [+0.01321, +0.03776]
19	Token minus WHOLE	PSNR	+0.40470 [+0.30712, +0.50909]
19	Token minus WHOLE	DISTS	-0.00447 [-0.00648, -0.00260]
19	Token minus WHOLE	LPIPS	-0.00822 [-0.01067, -0.00601]
19	Token minus WHOLE	DreamSim	-0.00331 [-0.00549, -0.00113]
19	Token minus WHOLE	CLIP-I	+0.00091 [-0.00623, +0.00705]
4	Token minus Quarter	PSNR	+0.17179 [+0.05088, +0.31324]
4	Token minus Quarter	DISTS	+0.00097 [-0.00384, +0.00561]
4	Token minus Quarter	LPIPS	-0.01085 [-0.01598, -0.00597]
4	Token minus Quarter	DreamSim	-0.00499 [-0.01620, +0.00392]
4	Token minus Quarter	CLIP-I	+0.00269 [-0.01127, +0.01503]
10	Token minus Quarter	PSNR	+0.12065 [+0.07247, +0.17387]
10	Token minus Quarter	DISTS	-0.00320 [-0.00624, -0.00044]
10	Token minus Quarter	LPIPS	-0.00467 [-0.00801, -0.00151]
10	Token minus Quarter	DreamSim	-0.00542 [-0.00960, -0.00162]
10	Token minus Quarter	CLIP-I	+0.01524 [+0.00610, +0.02431]
19	Token minus Quarter	PSNR	+0.17680 [+0.12165, +0.23538]
19	Token minus Quarter	DISTS	-0.00204 [-0.00369, -0.00062]
19	Token minus Quarter	LPIPS	-0.00438 [-0.00637, -0.00265]
19	Token minus Quarter	DreamSim	-0.00203 [-0.00364, -0.00032]
19	Token minus Quarter	CLIP-I	+0.00282 [-0.00206, +0.00751]
4	Token minus Ada	PSNR	+5.25672 [+4.60102, +5.93444]
4	Token minus Ada	DISTS	-0.06163 [-0.07486, -0.04864]
4	Token minus Ada	LPIPS	-0.19389 [-0.21446, -0.17418]
4	Token minus Ada	DreamSim	-0.13460 [-0.17473, -0.09885]
4	Token minus Ada	CLIP-I	+0.05372 [+0.02390, +0.08339]
10	Token minus Ada	PSNR	+1.20577 [+0.66352, +1.84723]
10	Token minus Ada	DISTS	+0.02214 [+0.01196, +0.03122]
10	Token minus Ada	LPIPS	-0.00238 [-0.01419, +0.00860]
10	Token minus Ada	DreamSim	+0.00321 [-0.01324, +0.01611]
10	Token minus Ada	CLIP-I	+0.00336 [-0.01758, +0.02570]
19	Token minus Ada	PSNR	+1.58860 [+1.39589, +1.78090]
19	Token minus Ada	DISTS	+0.01168 [+0.00768, +0.01569]
19	Token minus Ada	LPIPS	-0.02218 [-0.02645, -0.01836]
19	Token minus Ada	DreamSim	-0.00524 [-0.01259, +0.00326]
19	Token minus Ada	CLIP-I	+0.00012 [-0.01260, +0.01418]
4	Quarter minus WHOLE	PSNR	+0.39560 [+0.23628, +0.54342]
4	Quarter minus WHOLE	DISTS	-0.01208 [-0.01674, -0.00753]
4	Quarter minus WHOLE	LPIPS	-0.01894 [-0.02466, -0.01299]
4	Quarter minus WHOLE	DreamSim	-0.02760 [-0.03997, -0.01587]
4	Quarter minus WHOLE	CLIP-I	+0.01896 [+0.00878, +0.02989]
10	Quarter minus WHOLE	PSNR	+0.67356 [+0.40750, +0.95752]
10	Quarter minus WHOLE	DISTS	-0.01057 [-0.01587, -0.00576]
10	Quarter minus WHOLE	LPIPS	-0.02671 [-0.03701, -0.01667]
10	Quarter minus WHOLE	DreamSim	-0.01642 [-0.02618, -0.00721]
10	Quarter minus WHOLE	CLIP-I	+0.00930 [-0.00355, +0.02510]
19	Quarter minus WHOLE	PSNR	+0.22790 [+0.14025, +0.32419]
19	Quarter minus WHOLE	DISTS	-0.00243 [-0.00391, -0.00110]

SNR	Comparison	Metric	Difference [paired 95% CI]
19	Quarter minus WHOLE	LPIPS	-0.00383 [-0.00563, -0.00223]
19	Quarter minus WHOLE	DreamSim	-0.00127 [-0.00294, +0.00034]
19	Quarter minus WHOLE	CLIP-I	-0.00190 [-0.00760, +0.00222]
4	Quarter minus Ada	PSNR	+5.08493 [+4.42344, +5.78332]
4	Quarter minus Ada	DISTS	-0.06259 [-0.07638, -0.04927]
4	Quarter minus Ada	LPIPS	-0.18304 [-0.20325, -0.16354]
4	Quarter minus Ada	DreamSim	-0.12961 [-0.16912, -0.09369]
4	Quarter minus Ada	CLIP-I	+0.05103 [+0.02591, +0.07808]
10	Quarter minus Ada	PSNR	+1.08511 [+0.56422, +1.69685]
10	Quarter minus Ada	DISTS	+0.02533 [+0.01622, +0.03347]
10	Quarter minus Ada	LPIPS	+0.00229 [-0.00922, +0.01301]
10	Quarter minus Ada	DreamSim	+0.00864 [-0.00883, +0.02165]
10	Quarter minus Ada	CLIP-I	-0.01188 [-0.02758, +0.00382]
19	Quarter minus Ada	PSNR	+1.41180 [+1.22595, +1.60036]
19	Quarter minus Ada	DISTS	+0.01371 [+0.00969, +0.01776]
19	Quarter minus Ada	LPIPS	-0.01779 [-0.02225, -0.01377]
19	Quarter minus Ada	DreamSim	-0.00321 [-0.01010, +0.00507]
19	Quarter minus Ada	CLIP-I	-0.00269 [-0.01363, +0.00935]
4	WHOLE minus Ada	PSNR	+4.68933 [+4.03011, +5.37820]
4	WHOLE minus Ada	DISTS	-0.05052 [-0.06303, -0.03884]
4	WHOLE minus Ada	LPIPS	-0.16410 [-0.18559, -0.14359]
4	WHOLE minus Ada	DreamSim	-0.10201 [-0.14076, -0.06682]
4	WHOLE minus Ada	CLIP-I	+0.03207 [+0.00520, +0.05914]
10	WHOLE minus Ada	PSNR	+0.41155 [-0.15764, +1.06101]
10	WHOLE minus Ada	DISTS	+0.03590 [+0.02542, +0.04549]
10	WHOLE minus Ada	LPIPS	+0.02900 [+0.01505, +0.04273]
10	WHOLE minus Ada	DreamSim	+0.02506 [+0.00693, +0.04098]
10	WHOLE minus Ada	CLIP-I	-0.02118 [-0.04404, +0.00135]
19	WHOLE minus Ada	PSNR	+1.18390 [+1.02522, +1.34532]
19	WHOLE minus Ada	DISTS	+0.01615 [+0.01219, +0.01997]
19	WHOLE minus Ada	LPIPS	-0.01396 [-0.01862, -0.00960]
19	WHOLE minus Ada	DreamSim	-0.00194 [-0.00915, +0.00644]
19	WHOLE minus Ada	CLIP-I	-0.00079 [-0.01303, +0.01192]

SNR	Comparison	Metric	Better/tie/worse	Median gain
4	Token vs WHOLE	PSNR	23/0/1	+0.58584
4	Token vs WHOLE	DISTS	18/0/6	+0.01016
4	Token vs WHOLE	LPIPS	23/0/1	+0.03394
4	Token vs WHOLE	DreamSim	21/0/3	+0.02514
4	Token vs WHOLE	CLIP-I	20/0/4	+0.01863
10	Token vs WHOLE	PSNR	22/0/2	+0.77979
10	Token vs WHOLE	DISTS	18/0/6	+0.01544
10	Token vs WHOLE	LPIPS	21/0/3	+0.04019
10	Token vs WHOLE	DreamSim	19/0/5	+0.01697
10	Token vs WHOLE	CLIP-I	21/0/3	+0.01717
19	Token vs WHOLE	PSNR	24/0/0	+0.44240
19	Token vs WHOLE	DISTS	20/0/4	+0.00378
19	Token vs WHOLE	LPIPS	23/0/1	+0.00659
19	Token vs WHOLE	DreamSim	18/0/6	+0.00273
19	Token vs WHOLE	CLIP-I	16/0/8	+0.00226
4	Token vs Quarter	PSNR	14/0/10	+0.06004
4	Token vs Quarter	DISTS	13/0/11	+0.00089
4	Token vs Quarter	LPIPS	21/0/3	+0.00921
4	Token vs Quarter	DreamSim	12/0/12	+0.00097
4	Token vs Quarter	CLIP-I	15/0/9	+0.00541

SNR	Comparison	Metric	Better/tie/worse	Median gain
10	Token vs Quarter	PSNR	21/0/3	+0.08544
10	Token vs Quarter	DISTS	15/0/9	+0.00078
10	Token vs Quarter	LPIPS	18/0/6	+0.00318
10	Token vs Quarter	DreamSim	15/0/9	+0.00301
10	Token vs Quarter	CLIP-I	19/0/5	+0.01124
19	Token vs Quarter	PSNR	22/1/1	+0.14001
19	Token vs Quarter	DISTS	16/1/7	+0.00162
19	Token vs Quarter	LPIPS	21/1/2	+0.00274
19	Token vs Quarter	DreamSim	17/1/6	+0.00226
19	Token vs Quarter	CLIP-I	13/1/10	+0.00175
4	Token vs Ada	PSNR	24/0/0	+5.19071
4	Token vs Ada	DISTS	23/0/1	+0.05022
4	Token vs Ada	LPIPS	24/0/0	+0.19064
4	Token vs Ada	DreamSim	23/0/1	+0.10509
4	Token vs Ada	CLIP-I	19/0/5	+0.04465
10	Token vs Ada	PSNR	17/0/7	+1.01499
10	Token vs Ada	DISTS	3/0/21	-0.02454
10	Token vs Ada	LPIPS	13/0/11	+0.00181
10	Token vs Ada	DreamSim	7/0/17	-0.00732
10	Token vs Ada	CLIP-I	13/0/11	+0.00184
19	Token vs Ada	PSNR	24/0/0	+1.55558
19	Token vs Ada	DISTS	3/0/21	-0.01190
19	Token vs Ada	LPIPS	24/0/0	+0.02016
19	Token vs Ada	DreamSim	15/0/9	+0.00406
19	Token vs Ada	CLIP-I	8/0/16	-0.00586